BOUNDS FOR THE LENGTH OF FINITE PIERCE EXPANSIONS

JASON DAVIES

ABSTRACT. For integers $a > b > 0$, let $P(a,b)$ count the positive terms of the iteration $x \mapsto a \bmod x$, starting at b. We prove

$$P(a,b) \ll_\varepsilon a^{\beta_*+\varepsilon}, \qquad \beta_* = \frac{41301 - 3\sqrt{82849}}{137480} = 0.2941336553\ldots.$$

Positive integer Hankel determinants and related minors group short orbit segments into a small number of patterns. The fixed-dividend equations place the quotient increments for each pattern on affine curves. We improve the resulting integer-point count by adjoining further integer invariants, reducing their lattice coordinates, and combining an effective Hilbert irreducibility sieve with an exact polar identity. We also prove an elementary $3/10+\varepsilon$ bound and a bound for orbits with few rises in their quotient increments. The proofs, including the curve-counting input and the required effective Hilbert irreducibility theorem, are formalized in Lean 4 with Mathlib without proof holes or additional mathematical hypotheses. Shallit's conjectured $O((\log a)^2)$ bound remains open.

1. INTRODUCTION

Let $a > b > 0$ be integers. Put $b_0 = b$ and $b_{i+1} = a \bmod b_i$ while $b_i > 0$. The sequence decreases until it reaches 0; if $b_n = 0$ we write $P(a,b) = n$. For example, $P(35,22) = 7$, from $22, 13, 9, 8, 3, 2, 1, 0$. This is the length of the Pierce expansion of b/a [12, 13], since

$$\frac{b}{a} = \frac{1}{q_0} - \frac{1}{q_0 q_1} + \frac{1}{q_0 q_1 q_2} - \cdots, \qquad q_i = \lfloor a/b_i \rfloor.$$

Shallit [11] asked whether

$$P(a,b) = O\big((\log a)^2\big). \tag{1}$$

Shallit [13] proved $P(a,b) \ll a^{1/2}$ by "Archimedean" arguments. Erdős and Shallit [6] combined these with a divisor argument to obtain $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$, and showed that $\max_b P(a,b)/\log a$ does not tend to 0. Chase and Pandey [5] improved the upper exponent to $1/3 - 2/177$, using exponential sum estimates, and proved $\max_b P(a,b) \gg \log a/\log\log a$ for all large a. Baraskar and Vukusic [1] study the related sets of iterated remainders at fixed iteration depth. Shallit's talk pages [14] report a bound $n^{3/10}$, described as "LLM, unpublished so far". We have not seen that argument; our elementary bound below has the same exponent up to ε, and we make no claim about how the arguments are related.

Our main result is the following uniform bound.

Theorem 1.1. *For every $\varepsilon > 0$ there is $C_\varepsilon > 0$, independent of a, b, such that*

$$P(a,b) \le C_\varepsilon a^{\beta_*+\varepsilon} \qquad (0 < b < a), \qquad \beta_* = \frac{41301 - 3\sqrt{82849}}{137480}.$$

The fixed powers $(m,n) = (3,2)$ in the proof give the rational exponent $4317/14677$.

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Here $\beta_* = 0.2941336553\ldots$, while $4317/14677 = 0.2941336785\ldots$. The argument combines small integer invariants of the orbit with coefficient-uniform counting on curves. Its final improvement comes from a specialization sieve and a linear description of the curvature direction.

From integer invariants to curves. Write $k_i = q_{i+1} - q_i$ for the quotient increments and $d_i = b_i - b_{i+1}$ for the drops. Four consecutive terms x, y, z, w have drops d, e, f and two consecutive quotient increments u, v. Their curvature $D = dv - eu$ satisfies

$$Dz = df + ude + (u-1)e^2. \tag{2}$$

Thus D is a positive integer, although its defining terms largely cancel, and it is small when the drops are small. A six-term identity supplies a second positive integer L and controls the quotient by the small parameters of a single block (Section 5).

These integers belong to one hierarchy. Each tail carries a finite positive measure ν_i of mass b_i/a, and

$$I_r(i) = a \det\left(\int x^{j+l}\, d\nu_i\right)_{j,l<r}$$

is a positive integer when the tail has at least r positive terms. The first four invariants are b_i, d_i, D_i, L_i, in the nonterminal ranges of Theorem 6.2.

We count the T orbit terms in a remainder window $[A, 2A)$, then sum over dyadic A. A fixed proportion of starts have their next few increments bounded by $H \asymp a/(AT)$. The ten maximal minors of a 3×5 matrix of drops and increments, including $I_4(i), I_4(i+1), I_4(i+2)$, then have absolute value at most $X \asymp A^4H^6/a^3$. Their vector is the *pattern* of the start. The Plücker relations reduce the number of patterns to $a^{o(1)}X^5$ (Section 7).

For a fixed pattern, three nodes $s_j = q_{i+j} - q_i$, $j = 1, 2, 3$, determine the increments. The nodes satisfy a quadratic equation and, on retaining the fixed dividend a, a second equation of degree at most twelve. They therefore lie on a union of curves of total degree at most 24. We adjoin three integer curvature minors $W = (W_0, W_1, W_2)$ and the fifth Hankel invariant $J = I_5(i)$, whose sizes satisfy

$$|W_j| \ll S := A^3H^3/a^2, \qquad |J| \ll Y := A^5H^{10}/a^4.$$

The zeros and poles of these functions give lower bounds for the degree of the graph obtained by adjoining W_0^m, W_1^m, W_2^m, J^n, for fixed positive integers m, n. This makes the coefficient-uniform curve bound of Castryck, Cluckers, Dittmann and Nguyen [3] effective: an integral curve over $\mathbb{Q}$ of fixed degree δ in a box of size B has at most $B^{1/\delta+o(1)}$ integer points. Conics and components on which the reconstruction fails are counted separately.

The pattern also determines a rank-three lattice. If λ is the smallest sup norm of a nonzero lattice vector, a reduced basis gives integer gap and curvature coordinates of sizes $O(H/\lambda)$ and $O(S/\lambda^2)$. Averaged bounds for short lattice vectors improve the sum over patterns without changing the graph degree (Section 9).

The polar improvement. Two further facts give Theorem 1.1. First, an effective Hilbert irreducibility sieve forces the relevant curve component to have degree 24 outside a sparse set of patterns. Second, the exact polar identity identifies the projective direction $[W]$ with three affine-linear forms in the gaps. On the normalized curve, the common zeros of these forms, counted with multiplicity and residue degree, contribute at most four, outside a fixed proper algebraic exceptional locus. For the powered graph this gives the lower bound

$$\deg \Gamma \geq \frac{5n+3m}{2}\, 24 + m(24-4) = 60n + 56m.$$

The sieve and the geometric exceptional locus together discard at most $O_\eta(a^\eta X^{14/3})$ patterns, which are handled by the earlier curve estimate. Applying the curve bound in the reduced lattice coordinates and optimizing the fixed ratio n/m gives β_* (Section 10). This completes the route from the orbit to the main exponent.

Intermediate bounds. The two curve-counting stages before the polar improvement give useful bounds without the specialization sieve. Put

$$\gamma_* = \frac{471}{1559} - \frac{6\sqrt{105}}{7795} = 0.2942294\ldots, \qquad \alpha_* = \frac{1310 - \sqrt{655}}{4365} = 0.2942513\ldots.$$

Theorem 1.2 (Exterior lattice bound)**.** *For every $\varepsilon > 0$, $P(a,b) \ll_\varepsilon a^{\gamma_*+\varepsilon}$ uniformly for $0 < b < a$.*

Theorem 1.3 (Joint-invariant bound)**.** *For every $\varepsilon > 0$, $P(a,b) \ll_\varepsilon a^{\alpha_*+\varepsilon}$ uniformly for $0 < b < a$.*

These stages fit into the progression below. The first four bounds are formalized in the repository, but their proofs are not included here.

Exponent	Main input
23/77	I_4 as an integer label
11/37	a ternary quadratic and norm equations on slices
8/27	the $X^{5+o(1)}$ count of minor vectors
45/152	lattices of pivot gaps and their short vectors
α_*	adjoining curvature and rank-five invariants
γ_*	reducing the joint coordinates in a lattice basis

Fixed powers give the rational instances 128/435 and 2201/7480 of the joint-invariant bound, and 153/520 of the exterior lattice bound; see Corollaries 8.14 and 9.12. The repository also proves $P(a,b) \ll a^{3/10}\sqrt{\log\log 3a}$ conditional on the short-sum estimate of Nair and Tenenbaum [9] and the mean-value estimate of de la Bretèche and Tenenbaum [2] for the Erdős–Hooley function. These analytic estimates are explicit hypotheses; their proofs are not part of this formalization.

Elementary and structured bounds. The initial identities already yield an elementary improvement over the exponent 1/3.

Theorem 1.4. *For every $\varepsilon > 0$, $P(a,b) \ll_\varepsilon a^{3/10+\varepsilon}$ uniformly for $0 < b < a$.*

The proof uses only division equations and the divisor estimate $\tau(n) \ll_\eta n^\eta$. For a window of T remainders in $[A, 2A)$,

$$T(T-1) \le 4A\Delta, \tag{3}$$

$$T^6A^2 \ll a^3\Delta, \tag{4}$$

where $\Delta = \max_{1\le n\le 32a^7} \tau(n)$. The first is the Erdős–Shallit drop argument. For the second, there are few small triples (u, v, D), and a divisor argument bounds how often a triple can occur. Multiplying the second bound by the square of the first eliminates A and gives $T^{10} \ll a^3\Delta^3$. The six-step identity further gives

$$T^6A^2 \ll a^3, \tag{5}$$

without a divisor loss. This sharper estimate is used in the later arguments.

There is also a bound sensitive to the order of the increments. Let

$$R = 1 + \#\{i : 0 \le i \le P(a,b) - 3,\ k_i < k_{i+1}\}.$$

Theorem 1.5. *$P(a,b) \ll a^{1/4}\sqrt{R}$ uniformly for $0 < b < a$. In particular, nonincreasing quotient increments give $P(a,b) \ll a^{1/4}$.*

Curvature positivity makes the drops decrease between rises. The six-step identity limits repeated equal increments, and summing the resulting window bounds proves the theorem. This argument is independent of the curve-counting refinements.

Reading guide and formal verification. Sections 3–5 establish the elementary identities and bound. Sections 6–7 construct the integer invariants and their patterns. Section 8 builds the enlarged curves, Section 9 reduces their lattice coordinates, and Section 10 completes the main theorem. The independent rise bound is proved in Section 11. The conic count, graph-degree upper bound, and three-dimensional lattice reduction are proved in Appendices A, B and C.

All the unconditional results above, including their curve-counting and specialization inputs, are formalized in Lean 4 [7] with Mathlib [8]. Section 12 identifies the proof artifact and its verification scope. Section 13 explains the present 5/17 barrier; Shallit's logarithmic conjecture remains open.

2. Preliminaries

2.1. **Notation.** Throughout, $a > b > 0$ are integers, $b_0 = b$, $b_{i+1} = a \bmod b_i$, and $P = P(a,b)$, so that $a > b_0 > b_1 > \cdots > b_{P-1} > 0 = b_P$. Write $q_i = \lfloor a/b_i \rfloor$, so

$$a = q_i b_i + b_{i+1}, \qquad 0 \le b_{i+1} < b_i \qquad (0 \le i < P).$$

An index i of the orbit is also called a *state*; quantities "at state i" are computed from $b_i, b_{i+1}, \ldots$ and $q_i, q_{i+1}, \ldots$, and are defined whenever the orbit terms they use exist. Let $\tau(n)$ be the number of positive divisors of n, and let

$$\Delta = \Delta(a) = \max_{1 \le n \le 32a^7} \tau(n).$$

We write $F \ll_\eta G$ when $F \le C_\eta G$, with the constant uniform in the dividend and orbit. A factor denoted $a^{o(1)}$ is bounded by $C_\eta a^\eta$ for every fixed $\eta > 0$; such factors include the divisor losses below. We write $F \asymp G$ when both $F \ll G$ and $G \ll F$ hold. Many statements below carry explicit numerical constants; these are the constants of the Lean proof. Where a constant is large we state the bound with $\ll$ and record the constant in the proof.

Lemma 2.1 (Quotients increase)**.** *If $b_{i+1} > 0$ then $q_{i+1} \ge q_i + 1$.*

Proof. Since $b_{i+1} < b_i$ we have $(q_i+1)b_{i+1} \le q_i b_i + b_{i+1} = a$, so $q_{i+1} = \lfloor a/b_{i+1} \rfloor \ge q_i + 1$. □

Lemma 2.2 (Divisor bound)**.** *For every integer $k \ge 1$ there is C_k with $\tau(n)^k \le C_k\, n$ for all $n \ge 1$.*

Proof. This is standard: $\tau(n)^k/n = \prod_{p^\alpha \| n} (\alpha+1)^k/p^\alpha$. Each factor with $p \ge 2^k$ is at most $(\alpha+1)^k/2^{k\alpha} \le 1$. Each of the fewer than 2^k factors with $p < 2^k$ is at most $\sup_\alpha (\alpha+1)^k/2^\alpha < \infty$. □

2.2. **Window runs.**

Definition 2.3. Let $A \ge 1$ be an integer. A *window run* of length T is a sequence $x_0, x_1, \ldots, x_{T-1}$ of consecutive terms of the sequence (b_i), so $x_{j+1} = a \bmod x_j$, all lying in $[A, 2A)$.

In a window run, $x_0 > x_1 > \cdots > x_{T-1}$ and $x_j \le a$.

3. The curvature of three consecutive steps

Fix four consecutive terms

$$x = b_i, \quad y = b_{i+1} = a \bmod x, \quad z = b_{i+2} = a \bmod y, \quad w = b_{i+3} = a \bmod z,$$

with $x, y, z > 0$ (and $w \geq 0$). Write

$$q = \left\lfloor \frac{a}{x} \right\rfloor, \quad u = \left\lfloor \frac{a}{y} \right\rfloor - q, \quad v = \left\lfloor \frac{a}{z} \right\rfloor - \left\lfloor \frac{a}{y} \right\rfloor, \qquad d = x - y, \quad e = y - z, \quad f = z - w.$$

By Lemma 2.1, $u, v \geq 1$, and clearly $d, e, f \geq 1$. The division equations read

$$xq + y = a, \qquad y(q+u) + z = a, \qquad z(q+u+v) + w = a. \tag{6}$$

Definition 3.1. The *curvature* of the block is $D = dv - eu$.

Lemma 3.2 (Curvature identity)**.**

$$D\,z = df + ude + (u-1)e^2.$$

In particular $D \geq 1$.

Proof. Subtracting the first two equations of (6) gives $uy = qd + e$, hence $uz = uy - ue = qd + (1-u)e$. Subtracting the last two gives $vz = (q+u)e + f$. Therefore

$$(dv - eu)z = d\big((q+u)e + f\big) - e\big(qd + (1-u)e\big) = df + ude + (u-1)e^2.$$

The right-hand side is at least $df \geq 1$, and $z > 0$, so $D > 0$. □

Lemma 3.3 (The fiber equation)**.** *Put* $M(q) = uq^2 + (u^2+u+v)q + v$ *and* $N(q) = uq + u^2 + v$. *Then*

$$x\,M(q) = a\,N(q) + D.$$

Proof. From (6), $d = (q+1)x - a$ and $e = y - z = (q+u+1)y - a = a(q+u) - q(q+u+1)x$. Substituting these into $D = dv - eu$ gives $D = x\big(uq(q+u+1) + v(q+1)\big) - a\big(u(q+u) + v\big)$, which is the claim. □

Lemma 3.4 (Size of the parameters)**.** *If* $x \leq a$ *then*

$$u\,yx \leq a(d+e), \qquad v\,zy \leq a(e+f), \qquad D\,z \leq (u+1)(d+e+f)^2.$$

Proof. From $uy = qd + e$ we get $uyx = (qx)d + ex \leq ad + ae$, since $qx \leq a$ and $x \leq a$. Similarly $vz = (q+u)e + f$ with $(q+u)y \leq a$ and $y \leq a$ gives the second bound. For the third, by Lemma 3.2, $Dz = df + ude + (u-1)e^2 \leq df + u\,e(d+e) \leq (1+u)(d+e+f)^2$. □

4. Proof of Theorem 1.4

4.1. Few blocks share a triple (u, v, D).

Lemma 4.1 (Fiber count)**.** *Let* $a \geq 1$, $1 \leq u, v \leq a$ *and* $1 \leq D \leq 2a$. *Let* $M(q) = uq^2 + pq + v$ *with* $p = u^2 + u + v$, *and let* $\Lambda(q) = cq + \ell$ *with* $c = au$ *and* $\ell = a(u^2+v) + D$. *Then the set*

$$Q = \{q \in \mathbb{Z}_{\geq 0} : \ xM(q) = \Lambda(q) \text{ for some integer } x \geq 1\}$$

has at most $\Delta = \max_{1 \leq n \leq 32a^7} \tau(n)$ *elements.*

Proof. Put $U(q) = ucq + cp - u\ell$ and $R = u\ell^2 - cp\ell + c^2 v$. Expanding shows

$$c^2 M(q) - U(q)\Lambda(q) = R \tag{7}$$

identically in q. If $q \in Q$ with witness x, substituting $\Lambda(q) = xM(q)$ into (7) gives

$$R = M(q)\big(c^2 - U(q)x\big). \tag{8}$$

We bound the sizes first. From $u, v \le a$ and $D \le 2a$ we get $\ell \le 4a^3$, $p \le 3a^2$ and $c \le a^2$, so

$$|R| \le u\ell^2 + cp\ell + c^2 v \le 16a^7 + 12a^7 + a^5 \le 32a^7, \qquad c^2 \le a^4 \le 32a^7.$$

Case $R \neq 0$. By (8), $M(q)$ is a positive divisor of $|R|$. Since $u, p \ge 1$, the map $q \mapsto M(q)$ is strictly increasing on $q \ge 0$, hence injective. Therefore $|Q| \le \tau(|R|) \le \Delta$.

Case $R = 0$. Since $M(q) > 0$, (8) gives $U(q)x = c^2$. Here $c^2 > 0$ and $x \ge 1$, so $U(q)$ is a positive divisor of c^2. Since U is linear with positive leading coefficient uc, it is injective. Therefore $|Q| \le \tau(c^2) \le \Delta$. □

The case $R = 0$ cannot be dropped: it is the case in which M and Λ might share a factor.

4.2. **Counting visits to a window.** In this subsection $x_0, \dots, x_{T-1}$ is a window run in $[A, 2A)$, and $q_j = \lfloor a/x_j \rfloor$ are its quotients. We continue to write $\Delta = \max_{1 \le n \le 32a^7} \tau(n)$.

Proposition 4.2 (Drop bound; after Erdős–Shallit)**.** *If $T \ge 2$ then $T(T-1) \le 4A\Delta$.*

Proof. Let $h_j = x_j - x_{j+1} \ge 1$ for $0 \le j \le T-2$. The total drop is $\sum_j h_j = x_0 - x_{T-1} \le A - 1$. Thus fewer than $(T-1)/2$ drops exceed $2A/(T-1)$. The remaining set $\mathcal{S}$ has $2|\mathcal{S}| \ge T$.

For $j \in \mathcal{S}$ we have $1 \le h_j \le H := \lfloor 2A/(T-1) \rfloor$. Also $a + h_j = q_j x_j + x_{j+1} + x_j - x_{j+1} = (q_j + 1)x_j$, so $x_j \mid a + h_j$. Moreover $a + h_j \le 2a \le 32a^7$. Since $j \mapsto x_j$ is injective, at most $\tau(a + h) \le \Delta$ indices in $\mathcal{S}$ have $h_j = h$. Hence $|\mathcal{S}| \le H\Delta$, and

$$T(T-1) \le 2|\mathcal{S}|(T-1) \le 2\Delta\, H(T-1) \le 4A\Delta. \qquad \square$$

Proposition 4.3 (Curvature bound)**.** *If $T \ge 12$ then $T^6 A^2 \le 995328\, a^3 \Delta$.*

Proof. For $0 \le j \le T-4$ let $s_j = x_j - x_{j+3}$ be the total drop of the block $x_j, x_{j+1}, x_{j+2}, x_{j+3}$. By telescoping,

$$\sum_{j=0}^{T-4} s_j = (x_0 + x_1 + x_2) - (x_{T-3} + x_{T-2} + x_{T-1}) \le 3(2A - 1) - 3A = 3A - 3.$$

Call a block small if $s_j \le 12A/T$, and let $\mathcal{S}$ be the set of small blocks. Fewer than $T/4$ of the $T-3$ blocks fail this condition, so $|\mathcal{S}| \ge T/2$ when $T \ge 12$.

Fix $j \in \mathcal{S}$. Apply Section 3 to $x = x_j$, so that $y, z, w = x_{j+1}, x_{j+2}, x_{j+3}$, all in $[A, 2A)$. Write $s = s_j = d + e + f$. Lemma 3.4 gives $uA^2 \le uyx \le a(d+e) \le as$. Multiplying by T and using $sT \le 12A$ gives $uA^2 T \le 12aA$, that is,

$$u\,AT \le 12a, \qquad \text{and likewise} \qquad v\,AT \le 12a.$$

Also $DA \le Dz \le (u+1)s^2 \le 2us^2$. Multiplying by T^2 gives $DAT^2 \le 2u(sT)^2 \le 288uA^2$, so $DT^2 \le 288uA$, and then

$$D\,T^3 \le 288\,u\,AT \le 3456\,a.$$

Thus the triple (u, v, D) of a small block lies in $[1, U]^2 \times [1, D^*]$, where

$$U = \lfloor 12a/(AT) \rfloor \le a, \qquad D^* = \lfloor 3456a/T^3 \rfloor \le 2a.$$

The inequalities $U \le a$ and $D^* \le 2a$ hold because $AT \ge 12$ and $T^3 \ge 1728$.

Now fix such a triple and consider the small blocks that have it. By Lemma 3.3, each one gives a point $q = q_j = \lfloor a/x_j \rfloor$ of the set Q of Lemma 4.1. The map $j \mapsto q_j$ is injective on

these blocks: q_j determines $x_j = \Lambda(q_j)/M(q_j)$, and x_j determines j. By Lemma 4.1, at most Δ small blocks share a triple. Consequently

$$\frac{T}{2} \le |\mathcal{S}| \le U^2 D^* \Delta \le \left(\frac{12a}{AT}\right)^2 \frac{3456a}{T^3}\Delta,$$

which is the claimed bound. □

Corollary 4.4 (Window bound)**.** *If $T \ge 12$ then $T^{10} \le 63700992\, a^3\Delta^3$.*

Proof. The drop bound and $T \le 2(T-1)$ give $T^4 \le 64A^2\Delta^2$. Multiply by the curvature bound and cancel A^2. □

Remark 4.5. In terms of $B = a/A$, Propositions 4.2 and 4.3 say $T \ll a^{1/2+o(1)}B^{-1/2}$ and $T \ll a^{1/6+o(1)}B^{1/3}$. The first is better when $B > a^{2/5}$, and both equal $a^{3/10+o(1)}$ at $B = a^{2/5}$. Corollary 4.4 takes the product of the two bounds, so no case split is needed.

4.3. **Conclusion of the proof.** Let $0 < b < a$. For $0 \le k \le \log_2 a$ let T_k be the number of indices $i < P$ with $2^k \le b_i < 2^{k+1}$. Every b_i with $i < P$ lies in $[1, a)$, so $P = \sum_{0\le k\le \log_2 a} T_k$. Since (b_i) is decreasing, the indices counted by T_k are consecutive. So when $T_k > 0$ they form a window run in $[A, 2A)$ with $A = 2^k$.

For every $\eta > 0$, Lemma 2.2 gives $\Delta \ll_\eta a^\eta$, since its defining maximum is over $n \le 32a^7$. Taking $\eta = 5\varepsilon/3$ in Corollary 4.4 yields $T_k \ll_\varepsilon a^{3/10+\varepsilon/2}$; the windows with $T_k < 12$ satisfy the same estimate. There are $O(\log a) \ll_\varepsilon a^{\varepsilon/2}$ windows, so summing proves Theorem 1.4. □

5. Six consecutive steps

Fix six consecutive terms $x_0 > x_1 > \cdots > x_5 \ge 0$ of the sequence (b_i) with $x_4 > 0$, and write their quotients as

$$\lfloor a/x_0\rfloor = q, \quad \lfloor a/x_1\rfloor = q+u, \quad \lfloor a/x_2\rfloor = q+u+v,$$
$$\lfloor a/x_3\rfloor = q+u+v+w, \quad \lfloor a/x_4\rfloor = q+u+v+w+z.$$

(Unlike in Section 3, here w and z denote quotient increments.) By Lemma 2.1, $u, v, w, z \ge 1$. Let $f_k = x_k - x_{k+1} \ge 1$ be the drops, and let

$$D = f_0 v - f_1 u, \qquad E = f_1 w - f_2 v, \qquad F = f_2 z - f_3 w$$

be the curvatures of the blocks starting at x_0, x_1 and x_2. Write $\mathcal{E}_k$ for the division equation $x_k q_k + x_{k+1} - a = 0$, where $q_k = \lfloor a/x_k\rfloor$. Put

$$\sigma(u,v,w) = u^2 v + (u-1)v^2 + uw,$$
$$\chi(u,v,w,z) = \big(u^2 + (u-1)v\big)z + w\big(u^3 + uv(2u-1) + (u-1)v^2\big) + w^2\big((u-1)(u+v)+1\big),$$
$$L = (vw + w^2 + z)\,D - \big(u^2 + (u-1)v\big)E - w\,(wf_0 - uf_2).$$

Lemma 5.1 (The six-step identity)**.** (1) $\sigma(v,w,z)\,D - \sigma(u,v,w)\,E = v\,L$.
(2) $q\,w\,L = \chi(u,v,w,z)\,E + \sigma(u,v,w)\,F$.

Proof. Part (1) is a polynomial identity: since $\sigma(v,w,z) \equiv -w^2$ and $\sigma(u,v,w) \equiv uw \pmod v$, and $wD + uE = v(wf_0 - uf_2)$, the difference is divisible by v, and expanding gives L. Part (2) holds modulo the division equations. Explicitly, $qwL - \chi E - \sigma(u,v,w)F$ equals

$$w(v^2w + vw^2 + vz - w^2)\,\mathcal{E}_0 - w(u^2w + 2uvw + uw^2 + uz + v^2w + vw^2 - vw + vz - w^2)\,\mathcal{E}_1$$
$$+ w(u^2v + u^2w + uv^2 + 2uvw + uw^2 + uw + uz - v^2 - vw)\,\mathcal{E}_2 - w(u^2v + uv^2 + uw - v^2)\,\mathcal{E}_3.$$

This is an identity of polynomials in a, q, the x_k and the increments, checked by expansion. □

Corollary 5.2. *Suppose $q \geq 1$ and $u, v, w, z \leq m$. Then $E, F \geq 1$, $L \geq 1$, and*

$$q \leq 9m^3E + 3m^3F.$$

Proof. Lemma 3.2, applied to the blocks starting at x_1 and x_2, gives $E, F \geq 1$. Clearly $\chi > 0$ and $\sigma \geq 0$, so by Lemma 5.1(2) $qwL > 0$. Since L is an integer, $L \geq 1$ and $qw \leq \chi E + \sigma F$. Bounding each monomial gives $\chi \leq 9m^3w$ and $\sigma(u, v, w) \leq 3m^3w$. Dividing by w gives the claim. □

So the quotient q, which is about a/A, is controlled by the small parameters of a single block of six terms. No counting over many blocks is needed.

Proposition 5.3 (Six-step window bound)**.** *For a window run of length $T \geq 10$,*

$$T^6A^2 \leq 96 \cdot 10^6\, a^3.$$

Proof. For $0 \leq j \leq T-6$ let $s_j = x_j - x_{j+5}$. By telescoping, $\sum_{j \leq T-6} s_j \leq 5(2A-1) - 5A = 5A-5$. So some j has $s_j(T-5) \leq 5A-5$, and since $T \leq 2(T-5)$, we get $s := s_j$ with $sT \leq 10A$. Apply Corollary 5.2 to $x_j, \ldots, x_{j+5}$, all of which lie in $[A, 2A)$, with $m = \max(u, v, w, z)$.

By Lemma 3.4 at x_j, x_{j+1} and x_{j+2}, each increment satisfies (increment) $\cdot A^2 \leq as$. Multiplying by T gives $mAT \leq 10a$. By Lemma 3.4 again, $EA \leq Ex_{j+3} \leq (v+1)s^2 \leq 2ms^2$, and likewise $FA \leq 2ms^2$. Hence $qA \leq 24m^4s^2$. Also $a = qx_j + x_{j+1} < (q+1)x_j < 2A(q+1) \leq 4Aq$. Therefore

$$qA \cdot A^4T^6 \leq 24\,(mAT)^4(sT)^2 \leq 24 \cdot 10^6\, a^4A^2,$$

so $qA^3T^6 \leq 24 \cdot 10^6 a^4$, and $aA^2T^6 \leq 4qA^3T^6 \leq 96 \cdot 10^6\, a^4$. □

Remark 5.4. In terms of $B = a/A$, Proposition 5.3 reads $T \ll a^{1/6}B^{1/3}$. This is Proposition 4.3 without the factor $\Delta^{1/6}$. Combined with Proposition 4.2 as in Corollary 4.4, it gives $T^{10} \ll a^3\Delta^2$ and another proof of Theorem 1.4. The Lean proof of Theorem 1.4 uses Corollary 4.4.

6. The Hankel hierarchy

From now on we write $N = P(a, b)$, so that $b_0 > \cdots > b_{N-1} > 0 = b_N$, and $k_i = q_{i+1} - q_i$. For each remaining tail we construct a probability measure on its quotient differences. Reweighting by the reciprocal quotient will make its total mass equal to b_i/a.

Put $p_i(j) = q_{i+j} - q_i$ for $0 \leq j < N - i$; the symbols p_i, w_i are local to this construction. Define the weights backwards from $w_{N-1}(0) = 1$ by

$$w_i(0) = 1 - \sum_{j=0}^{N-i-2} \frac{w_{i+1}(j)}{p_i(j+1)}, \qquad w_i(j+1) = \frac{w_{i+1}(j)}{p_i(j+1)} \quad (0 \leq j < N-i-1).$$

Let μ_i be the measure with mass $w_i(j)$ at $p_i(j)$, and let

$$d\nu_i(x) = \frac{d\mu_i(x)}{q_i + x}, \qquad h_i(s) = \int x^s\, d\nu_i = \sum_{j=0}^{N-i-1} \frac{w_i(j)p_i(j)^s}{q_{i+j}}.$$

The recursion is chosen to reproduce the division equation $q_ih_i(0) + h_{i+1}(0) = 1$ for $i+1 < N$.

Lemma 6.1 (The moment model)**.** (1) *The weights are positive and sum to* 1.

(2) *$h_i(0) = b_i/a$, that is, $b_i/a = \mathbb{E}\big[1/(q_i + X_i)\big]$ for $X_i \sim \mu_i$; and $h_{N-1}(s) = 0$ for $s \geq 1$.*

(3) *$h_i(s+1) = \sum_{t \leq s} \binom{s}{t} k_i^{s-t} h_{i+1}(t)$ for $i + 1 < N$.*

(4) *$h_i(s) = (-q_i)^s b_i/a + z$ with $z \in \mathbb{Z}$.*

Proof. (1) Since $p_i(j+1) \geq 1$, the sum defining $w_i(0)$ is at most 1. It is strictly less: if $N-i-1 \geq 2$, the term $j=1$ has $p_i(2) \geq 2$; if $i = N-2$, the last gap satisfies $q_{N-1} \geq q_{N-2}+2$, because $b_{N-1} \mid a$ gives $q_{N-1} - q_{N-2} - 1 = q_{N-2}(b_{N-2} - b_{N-1})/b_{N-1} \geq 1$. (2) Since $p_i(j+1) = q_{i+j+1} - q_i$, the definitions give $q_i h_i(0) + h_{i+1}(0) = 1$, which is the normalized division equation; and $h_{N-1}(0) = 1/q_{N-1} = b_{N-1}/a$; since $\mu_{N-1} = \delta_0$, also $h_{N-1}(s) = 0$ for $s \geq 1$. (3) Use $p_i(j+1) = k_i + p_{i+1}(j)$ and the binomial theorem. (4) For $i = N-1$ this holds with $z = -(-q_{N-1})^s b_{N-1}/a = -(-1)^s q_{N-1}^{s-1}$ for $s \geq 1$, since $q_{N-1}b_{N-1} = a$. Then induct backwards on i using (3), $b_{i+1}/a = 1 - q_i b_i/a$ and $k_i - q_{i+1} = -q_i$. □

The weights also explain the role of the Newton polynomials below. Write $\mu_i(g) = \int g \, d\mu_i$. For $i+1 < N$ and any polynomial g, the zero atom contributes nothing to $\mu_i(xg)$, and $p_i(j+1)w_i(j+1) = w_{i+1}(j)$ gives

$$\mu_i(xg(x)) = \mu_{i+1}\big(g(k_i + x)\big).$$

Put $s_j = q_{i+j} - q_i$ and $F_j(x) = \prod_{0 \leq l < j}(x - s_l)$, with $F_0 = 1$. Iterating this identity gives the *shift rule*

$$\mu_i(F_j(x)g(x)) = \mu_{i+j}\big(g(x + s_j)\big) \qquad (i + j < N). \tag{9}$$

Indeed, each step removes the first factor of F_j and translates the remaining nodes by the next quotient gap. Thus multiplication by a Newton polynomial advances along the orbit; this is why this basis is useful for bounding its Hankel determinants.

Theorem 6.2 (Hankel determinants)**.** *For $1 \leq r \leq N-i$, the number $I_r(i) = a \det\big(h_i(j+l)\big)_{0 \leq j,l < r}$ is a positive integer. Moreover $I_1(i) = b_i$, $I_2(i) = d_i$ (if $i+1 < N$), $I_3(i) = D_i$ (if $i+3 < N$), and $I_4(i) = L_i$ (if $i+5 < N$), where D_i and L_i are the curvature and the six-step integer of the blocks starting at b_i.*

Proof. Integrality. By Lemma 6.1(4), $h_i(s+1) + q_i h_i(s) \in \mathbb{Z}$. Multiply the Hankel matrix on the left by the unit lower triangular matrix that adds q_i times row $j-1$ to row j. This makes rows $1, \ldots, r-1$ integral and leaves the determinant unchanged. Expand along row 0, whose entries are $(-q_i)^l b_i/a + (\text{integer})$; then $a \det$ is an integer.

Positivity. The Hankel matrix is the Gram matrix of $1, x, \ldots, x^{r-1}$ in $L^2(\nu_i)$. Its quadratic form is $\sum_j w_i(j) q_{i+j}^{-1} \big(\sum_l c_l p_i(j)^l\big)^2$. If this vanishes, the polynomial $\sum_l c_l x^l$ vanishes at the $N - i \geq r$ distinct points $p_i(j)$, so it is zero, by the invertibility of the Vandermonde matrix. So the matrix is positive definite.

The first four. First, $I_1(i) = a h_i(0) = b_i$. For $i+1 < N$, write $\beta_j = b_{i+j}/a$. Then $h_i(0) = \beta_0$, $h_i(1) = \beta_1$ and $h_i(2) = k_i\beta_1 + \beta_2$. Hence

$$I_2(i) = a\big(\beta_0(k_i\beta_1 + \beta_2) - \beta_1^2\big) = a(\beta_0 - \beta_1) = d_i,$$

using $q_i\beta_0 + \beta_1 = q_{i+1}\beta_1 + \beta_2 = 1$. For the larger determinants, Lemma 6.1(2),(3) makes the entries $h_i(s)$, $s \leq 2r-2$, explicit polynomials in $\beta_j = b_{i+j}/a$ and $k_i, \ldots, k_{i+2r-4}$; at the end of the orbit this uses $h_{N-1}(s) = 0$ for $s \geq 1$, that is, $b_N = 0$. Expanding and reducing by $q_{i+j}\beta_j + \beta_{j+1} = 1$ gives D_i and L_i in the stated ranges. The formal proof records polynomial certificates for these two reductions. □

The ranges $i+3 < N$ and $i+5 < N$ cannot be relaxed to all i at which D_i or L_i is defined. For $a = 926$ and $b = 229$ the orbit is $229, 10, 6, 2$, so $N = 4$, and $I_3(1) = 984$ while $D_1 = 988$.

Remark 6.3. In these index ranges, the positivity of D and L, proved earlier by explicit identities, is an instance of the positivity of Gram determinants. The integrality of L (Lemma 5.1(1)) is an instance of the integrality of I_4. A priori $I_r(i)$ involves the moments $h_i(s)$ for $s \leq 2r-2$,

and hence $b_i, \dots, b_{i+2r-2}$. In fact $I_3(i)$ depends only on b_i, b_{i+1}, b_{i+2} and q_i, q_{i+1}, q_{i+2}. Similarly $I_4(i)$ depends only on $b_i, \dots, b_{i+3}$ and $q_i, \dots, q_{i+4}$: the later remainders cancel.

7. Patterns

In this section we develop the patterns used in Section 8, together with the bounds on their size and number. We keep the notation of Section 6: $b_0 > b_1 > \cdots > b_{N-1} > 0$ is the positive part of the orbit, $N = P(a, b)$, and $q_i = \lfloor a/b_i \rfloor$.

7.1. **Heuristics.** The exponent $3/10$ of Theorem 1.4 comes from a single balanced configuration: $T \asymp a^{3/10}$ remainders in a window $[A, 2A)$ with $A \asymp a^{3/5}$, quotients $q \asymp a^{2/5}$ and typical increments $k \asymp a^{1/10}$. At this scale the four-term Hankel invariant $I_4(i) = L(i)$ has size $\asymp aH^6/q^4 \asymp 1$, where H bounds the increments. The idea is that a positive integer of bounded size carries information. The labels $I_4(i), I_4(i+1), I_4(i+2)$, together with four further integers of the same size, form a *pattern* (made precise in Section 7.6). States with a given pattern satisfy two linear equations and one quadratic equation in five consecutive increments (Lemma 7.1), and the quotient q_i is a root of a quartic fixed by the pattern and the increments (Lemma 7.3). So the problem becomes one of counting patterns, and the states belonging to each pattern. Counting the latter with these equations alone already improves on $3/10$ (see the intermediate bounds in Section 1). Section 8 instead keeps the fixed dividend a in the equations.

7.2. **The pattern matrix.** Fix a state i with $i + 6 < N$ and write, for $m \geq 0$,

$$k_m = q_{i+m+1} - q_{i+m}, \qquad d_m = b_{i+m} - b_{i+m+1},$$
$$s_m = k_0 + \cdots + k_{m-1}, \qquad r_m = k_m(s_m + s_{m+1}) + k_{m+1}.$$

All indices here are local to the chosen state, and the definitions apply whenever the indicated orbit entries exist; the empty sum gives $s_0 = 0$. So $s_m = q_{i+m} - q_i$, $r_0 = k_0^2 + k_1$, $r_1 = k_1(2k_0 + k_1) + k_2$, and so on. Consider the 3×4 integer matrix

$$\mathcal{M}_i = \begin{pmatrix} d_0 & d_1 & d_2 & d_3 \\ k_0 & k_1 & k_2 & k_3 \\ r_0 & r_1 & r_2 & r_3 \end{pmatrix},$$

and let $U_m(i) = (-1)^{m+1} \det(\mathcal{M}_i \text{ with column } m \text{ deleted})$ for $0 \leq m \leq 3$.

Lemma 7.1 (Cofactor relations)**.** (1) *$U_3(i) = L(i) = I_4(i)$ and $U_0(i) = -L(i+1) = -I_4(i+1)$, where L is the six-step integer of Lemma 5.1.*
(2) *$\sum_m U_m(i)\, d_m = \sum_m U_m(i)\, k_m = \sum_m U_m(i)\, r_m = 0$.*

Proof. (2) is the Laplace expansion of a 4×4 determinant with a repeated row. For (1), the identity $\det(\text{columns } 0, 1, 2) = L(i)$ holds as a polynomial identity in $d_0, d_1, d_2, k_0, \dots, k_3$, by direct expansion; and $L(i) = I_4(i)$ by Theorem 6.2. Passing from i to $i + 1$ shifts the d- and k-rows by one column, and, since $s_m(i+1) = s_{m+1}(i) - k_0$, it replaces r_m by $r_{m+1} - 2k_0k_{m+1}$. This row operation does not change the determinant, so $\det(\text{columns } 1, 2, 3) = L(i+1)$. □

For example, the positive part of the orbit with $(a, b) = (95, 61)$ is $61, 34, 27, 14, 11, 7, 4, 3, 2, 1$. Its first pattern matrix and cofactor vector are

$$\mathcal{M}_0 = \begin{pmatrix} 27 & 7 & 13 & 3 \\ 1 & 1 & 3 & 2 \\ 2 & 6 & 23 & 29 \end{pmatrix}, \qquad U(0) = (-81, 833, -296, 68).$$

Thus $L(0) = 68$, $L(1) = 81$, and each row of $\mathcal{M}_0$ is orthogonal to $U(0)$.

Remark 7.2. The vector $C_m = U_m/U_3$ has a meaning: $\pi_i = \sum_{m\le 3} C_m(F_{m+1} - F_m)$, with Newton polynomials $F_m(x) = \prod_{l<m}(x - s_l)$, is the monic orthogonal polynomial of degree four for the measure ν_i of Section 6. The three rows of $\mathcal{M}_i$ encode $\nu_i(\pi_i) = 0$ and $\mu_i(x\pi_i) = \mu_i(x^2\pi_i) = 0$, and $\nu_i(xf) = \mu_i(f) - q_i\nu_i(f)$ turns these into $\nu_i(x^j\pi_i) = 0$ for $j \le 3$. In particular $U_m(i)/I_4(i)$ are tail sums of the Newton coefficients of π_i: writing $\pi_i = \sum_{l\le 4}\gamma_l F_l$, one has $U_m(i)/I_4(i) = \gamma_{m+1} + \cdots + \gamma_4$, and $\gamma_0 = -C_0 = I_4(i+1)/I_4(i)$. This interpretation is used in Section 8.

The *pattern integers* of i are the seven integers

$$\theta(i) = \big(L(i),\, L(i+1),\, L(i+2),\, U_1(i),\, U_2(i),\, U_1(i+1),\, U_2(i+1)\big),$$

and we write $\theta(i) = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$. By Lemma 7.1, $\theta(i)$ determines $U(i) = (-N_1, u_1, u_2, N_0)$ and $U(i+1) = (-N_2, v_1, v_2, N_1)$ completely, and the k-relations at i and $i+1$ read

$$N_1k_0 = u_1k_1 + u_2k_2 + N_0k_3, \qquad N_1k_4 = N_2k_1 - v_1k_2 - v_2k_3.$$

7.3. **Two identities.**

Lemma 7.3 (The normalization identity)**.** *Let π_i be the monic orthogonal polynomial of degree four for ν_i, as in the remark above, and let $H_4 = \det\big(\mu_i(x^{j+l})\big)_{j,l<4}$. Then H_4 is a polynomial in $k_0, \dots, k_4$ with 43 terms, independent of i, and*

$$I_4(i)\,\pi_i(-q_i) = a\,H_4(k_0, \dots, k_4).$$

As a polynomial in q, $I_4(i)\pi_i(-q)$ has degree four and leading coefficient $I_4(i) = L(i) \ge 1$.

Proof. Since $F_{m+1} - F_m = F_m(x)(x - s_m - 1)$, the remark gives

$$I_4(i)\pi_i(x) = \sum_{m=0}^{3} U_m(i)\,F_m(x)\,(x - s_m - 1).$$

The division equations read $b_{i+m+1} = a - (q_i + s_m)b_{i+m}$. Use them to eliminate $b_{i+1}, \dots, b_{i+4}$ from both sides. The difference becomes the zero polynomial in $a, q_i, b_i, k_0, \dots, k_4$; explicit certificates are in the Lean proof. Conceptually, the identity follows from the determinant formula for π_i, since $\nu_i(x^{s+1}) + q_i\nu_i(x^s) = \mu_i(x^s)$. The leading coefficient is that of the monic π_i. □

Lemma 7.4 (Labels)**.** *Let $i + 6 < N$ and put*

$$G(k_0, \dots, k_4) = (k_0 + 1)\big(k_1s_2 + s_2 + k_1 + k_2 + 1\big)\big(k_2(k_1 + k_2)s_3 + e_3\big),$$

where $e_3 = s_3(k_1 + k_2) + s_3k_2 + (k_1 + k_2)k_2 + (k_0 + 2k_1 + 3k_2)(k_3 + 1) + k_3^2 + 2k_3 + k_4 + 1$. Then $L(i)\,q_i^4 \le a\,G(k_0, \dots, k_4)$. If moreover $b_i < 2A$, then $L(i)\,a^3 \le 256A^4\,G(k_0, \dots, k_4)$.

Proof. The Newton polynomials $F_0, \dots, F_3$ are monic of degrees $0, \dots, 3$. So the matrix $\big(\nu_i(F_jF_l)\big)$ is $\mathcal{U}\mathcal{H}\mathcal{U}^{\mathsf{T}}$, where $\mathcal{H}$ is the Hankel matrix of ν_i and $\mathcal{U}$ is unit lower triangular. Hence it has determinant $I_4(i)/a$, and it is positive definite, as in the proof of Theorem 6.2. By Hadamard's inequality,

$$\frac{I_4(i)}{a} \le \prod_{j<4} \nu_i(F_j^2) \le \prod_{j<4} \frac{\mu_i(F_j^2)}{q_i},$$

since $\nu_i = \mu_i/(q_i + x) \le \mu_i/q_i$ on the nonnegative support. By the shift rule (9) and the moments $\mu_{i+m}(1) = \mu_{i+m}(x) = 1$, $\mu_{i+m}(x^2) = k_m + 1$, $\mu_{i+m}(x^3) = k_m^2 + 2k_m + k_{m+1} + 1$, the product equals G. For the second claim, $a = q_ib_i + b_{i+1} < 2A(q_i + 1) \le 4Aq_i$, so $L(i)a^4 \le 256A^4L(i)q_i^4 \le 256A^4aG$. □

7.4. **Signs.** Throughout this subsection the pattern and gaps arise from the same state, so the cofactor relations of Lemma 7.1 remain in force. Besides the polynomial σ of Section 5, put $\omega(x,y,z,w) = x^2z + (2x-1)yz + xz^2 + xw$. Both σ and ω are positive at positive integers.

Lemma 7.5 (Signs)**.** *If $N_0, N_1 \geq 1$ and $k_0, \dots, k_4 \geq 1$, then $u_1 > 0 > u_2$.*

Proof. Solving the k- and r-relations of Lemma 7.1 for (u_1, u_2) by Cramer's rule gives the exact identities

$$u_1\sigma(k_1,k_2,k_3) = N_1\omega(k_0,k_1,k_2,k_3) + N_0\sigma(k_2,k_3,k_4),$$
$$-u_2\sigma(k_1,k_2,k_3) = N_1\sigma(k_0,k_1,k_2) + N_0\omega(k_1,k_2,k_3,k_4),$$

since the Cramer determinant is $k_1r_2 - k_2r_1 = \sigma(k_1,k_2,k_3)$. □

7.5. **Good starts.**

Lemma 7.6 (Good starts)**.** *Let $b_s, \dots, b_{s+T-1}$ be a window run in $[A, 2A)$ with $T \geq 36$, and put $H = \lfloor 28a/(AT) \rfloor + 1$. Call a start i* good *if $s \leq i$, $i+9 \leq s+T$ and $q_{j+1} - q_j \leq H$ for $i \leq j \leq i+6$, that is, $k_0, \dots, k_6 \leq H$ at state i. Then at least $T/2$ starts are good, and every good start satisfies $i+8 < N$ and $b_i, b_{i+1}, b_{i+2} < 2A$.*

Proof. Let $\mathcal{B}$ be the set of $j \in [s, s+T-1)$ with $q_{j+1} - q_j > H$. Then $|\mathcal{B}|(H+1) \leq q_{s+T-1} - q_s \leq a/A$. Since $H \cdot AT > 28a$, this gives $28|\mathcal{B}| \leq T$. Each $j \in \mathcal{B}$ spoils at most seven starts, so at least $T - 9 - 7|\mathcal{B}| \geq T/2$ starts are good, as $T \geq 36$. The last assertion is immediate, since the states $i, \dots, i+8$ lie in the window. □

7.6. **The full minor vector.** The seven pattern integers are not independent: they are coordinates of a single vector of 3×3 minors. If all ten minors are at most X in absolute value, there are only $X^{5+o(1)}$ such vectors (Proposition 7.9), rather than the X^7 choices of seven independent integers. We therefore define the *pattern* of a state i to be the vector $p(i)$ of the ten 3×3 minors of the matrix $\widehat{\mathcal{M}}_i$ below; it determines the pattern integers $\theta(i)$.

Let $\widehat{\mathcal{M}}_i$ be the 3×5 matrix with rows $(d_0, \dots, d_4)$, $(k_0, \dots, k_4)$ and $(r_0, \dots, r_4)$, and let $p_{c_1c_2c_3}$ be its minor on columns $c_1 < c_2 < c_3$. Expanding the cofactors above gives

$$\theta(i) = (p_{012},\, p_{123},\, p_{234},\, p_{023},\, -p_{013},\, p_{134},\, -p_{124}).$$

Lemma 7.7 (Expansion)**.** *Let $q = q_i$ and $s_j = q_{i+j} - q$. If $i + j + 3 < N$ then*

$$\frac{b_{i+j}}{a} = \frac{1}{q} - \frac{1+s_j}{q^2} + \frac{s_j^2 + 2s_j + k_j + 1}{q^3} - e_j, \qquad 0 \leq e_j \leq \frac{\mu_{i+j}\big((x+s_j)^3\big)}{q^4}.$$

Proof. Apply μ_{i+j} to $\frac{1}{q+y} = \frac{1}{q} - \frac{y}{q^2} + \frac{y^2}{q^3} - \frac{y^3}{q^3(q+y)}$ with $y = s_j + x \geq 0$, using $b_{i+j}/a = \nu_{i+j}(1) = \mu_{i+j}\big(1/(q+s_j+x)\big)$, $\mu(1) = \mu(x) = 1$ and $\mu(x^2) = k+1$. □

Proposition 7.8 (All minors)**.** *Let $i+8 < N$, $b_i < 2A$ and $k_0, \dots, k_6 \leq H$ at state i. Then every minor of $\widehat{\mathcal{M}}_i$ satisfies $|p|\, a^3 \leq 3609600\, A^4H^6$.*

Proof. Subtracting the expansions of Lemma 7.7 for j and $j+1$ gives $d_j/a = k_j/q^2 - (r_j + k_j)/q^3 + e_{j+1} - e_j$. So replacing the first row by $d'_j = d_j - a(q^{-2} - q^{-3})k_j + aq^{-3}r_j = a(e_{j+1} - e_j)$ changes no minor. Now $|d'_j| \leq 235aH^3/q^4$, $|k_j| \leq H$ and $|r_j| \leq 10H^2$, so $|p| \leq 14100\, aH^6/q^4$. Finally $a \leq 4Aq$. □

7.7. **Counting minor vectors.**

Proposition 7.9 (Minor vectors)**.** *The number of vectors of minors of integer* 3×5 *matrices with all coordinates in* $[-X, X]$ *is at most* $1 + 20 \sum_{\kappa \le X} 729\, \kappa^4 \tau(\kappa) \ll X^{5+o(1)}$.

Proof. For a nonzero vector p choose a pivot π with $\kappa = |p_\pi|$ maximal, and let c_4, c_5 be the two columns outside π. Let u_r (v_r) be the minor with the r-th pivot column replaced by c_4 (c_5), with the sign of the column permutation. By Cramer's rule and the Plücker relations, the remaining three minors satisfy $p_\pi\, p_{w_r} = \pm(u \times v)_r$ (thirty polynomial identities, checked symbolically). So p is determined by $(\pi, \operatorname{sgn} p_\pi, \kappa, u, v)$, where $|u_r|, |v_r| \le \kappa$ and κ divides every coordinate of $u \times v$.

Fix $\kappa \ge 1$ and u, and let $g = \gcd(\kappa, u_1, u_2, u_3)$, $\ell = \kappa/g$, $u = g u_0$. By Bézout there are $z \in \mathbb{Z}^3$ and $t \in \mathbb{Z}$ with $z \cdot u_0 + t\ell = 1$. If $\kappa \mid u \times v$, then $\ell \mid u_0 \times v$, and the identity $v - (z \cdot v)u_0 = t\ell\, v + z \times (v \times u_0)$ shows $v = \mu u_0 + \ell y$ with $\mu = z \cdot v$ and $y \in \mathbb{Z}^3$. Hence v lies in one of ℓ classes modulo ℓ, each meeting $[-\kappa, \kappa]^3$ in at most $(3g)^3$ points: at most $27\kappa g^2$ choices. Summing over u, grouped by $g = e \mid \kappa$, gives $\sum_u g^2 \le \sum_{e \mid \kappa} e^2 (3\kappa/e)^3 \le 27\kappa^3 \tau(\kappa)$, so at most $729\kappa^4\tau(\kappa)$ pairs (u, v). There are ten pivots and two signs. □

8. Joint integer invariants and the optimized bound

We now prove Theorem 1.3. Fix a window run in $[A, 2A)$ with $T \ge 36$ terms. Throughout this section constants may depend on fixed positive integers m, n (the powers of Section 8.9) and on a loss $\eta > 0$, but never on the dividend, the orbit or the pattern. The argument has six steps.

(1) For a fixed pattern, the nodes of every good start lie on at most 24 curves (Section 8.1).
(2) On these curves the quotient q_i and the curvature $I_3(i)$ are rational functions of the nodes. Components on which the reconstruction identities fail contribute $O(1)$ starts (Section 8.2).
(3) The directions at infinity of the curves are controlled by the Newton coefficients of the pattern. This makes nonexceptional components have even degree (Section 8.3), shows that exceptional patterns are few, and excludes lines (Section 8.4).
(4) The reconstructed functions, and hence two families of new integer coordinates W and J, have prescribed orders at infinity (Sections 8.5 and 8.6).
(5) Each nonexceptional conic contributes $a^{o(1)}$ starts (Section 8.7).
(6) On the remaining components, adjoining W_j^m and J^n raises the degree. The curve-counting theorem and a balancing of heights then give the exponent (Sections 8.8 and 8.9).

Notation. In this section

- $\theta = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$ are the pattern integers of Section 7, with $N_0 = I_4(i)$, and bars denote their normalized versions (10);
- s_j are the nodes, k_j the gaps, P_4, R_4 the two orthogonal quartics, Ψ the quartic norm and f the pattern quadratic;
- $D_{jl} = d_j k_l - d_l k_j$ are the 2×2 minors of drops and gaps, so that $D_{01} = I_3(i)$ is the curvature of Section 3;
- Q and E_0 are the reconstructed quotient and curvature, $W = (W_0, W_1, W_2)$ and J the joint coordinates, and S, Y their height bounds;
- e is a multiplicity at infinity, d the degree of a component, δ the degree of a graph and $\delta_{\min} = 10n + 7m$.

8.1. **Good starts and the pattern curve.** Put

$$H = \left\lfloor \frac{28a}{AT} \right\rfloor + 1, \qquad X = \left\lfloor \frac{3609600A^4H^6}{a^3} \right\rfloor.$$

Let the good starts be as in Lemma 7.6; at least $T/2$ starts are good. By Proposition 7.8, the minor vector of a good start has all coordinates at most X, and $X \geq 1$ because its coordinate $p_{012} = L(i)$ is a positive integer. The pattern of a good start (Section 7.6) therefore has height at most X, and it determines $\theta(i)$. By Proposition 7.9, there are $O_\eta(a^\eta X^5)$ patterns.

The *nodes* of state i are $s_j = q_{i+j} - q_i$. They are the support points of the measure μ_i of Section 6, and the interpolation nodes of the Newton polynomials of Section 7. The three *free nodes* s_1, s_2, s_3 of a good start are integers in $[0, 3H]$. Distinct good starts with the same pattern have distinct free-node triples (Lemma 8.2 below).

For the pattern integers θ, put

$$(\bar N_1, \bar u_1, \bar u_2) = \frac{(N_1, u_1, u_2)}{N_0}, \qquad (\bar N_2, \bar v_1, \bar v_2) = \frac{(N_2, v_1, v_2)}{N_1}. \tag{10}$$

By Lemma 7.5, $\bar N_1 > 0$, $\bar u_1 > 0$ and $\bar u_2 < 0$. Treat the free nodes s_1, s_2, s_3 as coordinates and define

$$s_0 = 0, \quad s_4 = (\bar N_1 + \bar u_1)s_1 - (\bar u_1 - \bar u_2)s_2 - (\bar u_2 - 1)s_3, \quad k_j = s_{j+1} - s_j \quad (0 \leq j \leq 3),$$
$$k_4 = \bar N_2 k_1 - \bar v_1 k_2 - \bar v_2 k_3, \qquad s_5 = s_4 + k_4.$$

At a good start these are the actual nodes and gaps, by the two k-relations of Lemma 7.1; the first is equivalent to $\bar N_1 k_0 = \bar u_1 k_1 + \bar u_2 k_2 + k_3$. Let

$$\gamma = (\bar N_1,\ -\bar N_1 - \bar u_1,\ \bar u_1 - \bar u_2,\ \bar u_2 - 1,\ 1),$$

and let γ' be the same expression in $\bar N_2, \bar v_1, \bar v_2$. Put

$$P_4(x) = \sum_{j=0}^{4} \gamma_j \prod_{\ell=0}^{j-1} (x - s_\ell), \qquad R_4(x) = \sum_{j=0}^{4} \gamma'_j \prod_{\ell=1}^{j} (x - s_\ell).$$

By the remark after Lemma 7.1, at a good start P_4 is the monic orthogonal polynomial π_i of degree four for ν_i, and $R_4(x) = \pi_{i+1}(x - k_0)$ is the corresponding polynomial of the next state, written in the variable of the current state. Put $Z = xR_4$ and $f = \sum_{j=0}^{4} \gamma_j (s_j^2 + k_j)$, the *pattern quadratic*, and let

$$\Psi = \sum_{j=0}^{4} \gamma_j \big(s_j^3 + 3s_j^2 + 3s_j(1 + k_j) + (1 + k_j)^2\big) + \sum_{j=0}^{3} \gamma_j k_{j+1}$$
$$+ \bar N_2 r_1 - \bar v_1 r_2 - \bar v_2 r_3 - k_4(2s_4 + k_4),$$

where $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$ for $j = 1, 2, 3$. This is a cubic polynomial in the free nodes, the *quartic norm*.

Degrees throughout this section mean the leading coefficient of the eventual affine Hilbert function, equivalently the degree of the projective closure. "Integral" means that the defining ideal is prime over $\mathbb{Q}$; it does not assert geometric irreducibility or a rational parametrization.

Lemma 8.1 (The pattern curve)**.** (1) *At a good start i, $\Psi = \nu_i(P_4^2) > 0$, and the free nodes satisfy*

$$f = 0, \qquad N_0 \operatorname{Res}(P_4, Z) = a\Psi^4. \tag{11}$$

(2) *For every pattern with $\bar N_1, \bar u_1 > 0 > \bar u_2$, the two equations* (11) *have degrees two and at most twelve and no common factor. Their common zero set in $\mathbb{A}^3$ is the union of at most twenty-four integral affine curves over $\mathbb{Q}$, of total degree at most twenty-four.*

Proof. (1) Since $r_m = s_{m+1}^2 - s_m^2 + k_{m+1}$, summation by parts and the k-relation at state i show that $N_0 f$ is the r-relation $\sum_m U_m(i) r_m = 0$ of Lemma 7.1. Next, the shift rule $\mu_i\big(g\prod_{\ell<j}(x - s_\ell)\big) = \mu_{i+j}\big(g(x+s_j)\big)$ and the moments listed in the proof of Lemma 7.4 give $\mu_i(x^3 P_4) = \sum_j \gamma_j\big(s_j^3 + 3s_j^2 + 3s_j(1+k_j) + (1+k_j)^2 + k_{j+1}\big)$. Eliminating k_5 with the r-relation at state $i+1$ turns this into Ψ. Since $\nu_i(xg) = \mu_i(g) - q_i\nu_i(g)$ and P_4 is orthogonal to x^3, $\nu_i(P_4^2) = \nu_i(x^4 P_4) = \mu_i(x^3 P_4) = \Psi$, and this is positive as in the proof of Theorem 6.2.

For the resultant, Lemma 6.1(3) gives $\nu_i\big(xg(x)\big) = \nu_{i+1}\big(g(x+k_0)\big)$. Hence, for $j < 4$, $\nu_i(x^j Z) = \nu_{i+1}\big((x+k_0)^j \pi_{i+1}(x)\big) = 0$. So the monic quintic Z differs from the monic orthogonal polynomial $\pi^{(5)}$ of degree five for ν_i by a multiple of P_4, and $\operatorname{Res}(P_4, Z) = \operatorname{Res}(P_4, \pi^{(5)})$. Let n_j be the squared ν_i-norm of the monic orthogonal polynomial of degree j. The three-term recurrence, whose coefficients are $\beta_j = n_j/n_{j-1}$, gives

$$\operatorname{Res}(P_4, \pi^{(5)}) = \beta_1\beta_2^2\beta_3^3\beta_4^4 = \frac{n_4^4}{n_0 n_1 n_2 n_3} = \frac{\Psi^4}{\det\big(\nu_i(x^{r+s})\big)_{r,s<4}} = \frac{a\Psi^4}{I_4(i)}.$$

Since $N_0 = I_4(i)$, this is the second equation.

(2) The degree bounds are immediate, since the resultant has total degree at most ten in the nodes. Note that $\gamma_1 < 0 < \gamma_2$, $\gamma_3 = \bar u_2 - 1 < -1$ and $\gamma_0, \gamma_4 > 0$. Since $s_0 = 0$ and $\sum_j \gamma_j s_j = 0$, the quadratic part of f is

$$f_2 = \gamma_1 s_1^2 + \gamma_2 s_2^2 + \gamma_3 s_3^2 + (\gamma_1 s_1 + \gamma_2 s_2 + \gamma_3 s_3)^2,$$

whose Hessian determinant is $8\gamma_1\gamma_2\gamma_3(1+\gamma_1+\gamma_2+\gamma_3) = -8\gamma_0\gamma_1\gamma_2\gamma_3 \neq 0$. A ternary quadratic form of rank three is not a product of linear forms, so f_2, and hence f, is irreducible. The degree-twelve part of the second equation is $-ac^4$ with $c = \sum_j \gamma_j s_j^3$. If f divided the second equation, the irreducible form f_2 would divide c^4, hence c. Divide c by f_2 as polynomials in s_3; the s_3^2 coefficient of f_2 is $\gamma_3(1+\gamma_3) \neq 0$. The coefficients of the remainder vanish only if $\gamma_1\gamma_2\gamma_3^2(\gamma_3^2 - 1) = 0$, which the signs above exclude. So the two equations have no common factor, their common zero set is a curve, and by Bézout's theorem its components over $\mathbb{Q}$ have total degree at most $2 \cdot 12$. □

8.2. **Reconstruction and retained components.** Two further identities distinguish the components on which the later rational functions have the required behavior. We give their definitions before stating the component bounds.

Form the 8×9 matrix of coefficients of $P_4, xP_4, x^2P_4, x^3P_4, Z, xZ, x^2Z, x^3Z$ in the basis $1, x, \dots, x^8$. Let c_j be $(-1)^j$ times the minor obtained by deleting column j, and define

$$Q = \frac{c_1 - c_2}{c_1 - c_0}, \qquad E_0 = a\,\frac{\det(c_{r+s})_{0\le r,s<3}(c_1 - c_0)^3}{(c_1^2 - c_0 c_2)^3}.$$

At a good start the matrix annihilates the moment vector $(h_i(0), \dots, h_i(8))$ and has rank eight. Indeed, the resultant in (11) is positive, so P_4 and Z are coprime. The identities $q_i h_i(0) + h_i(1) = q_i h_i(1) + h_i(2) = 1$, together with $h_i(0)h_i(2) > h_i(1)^2$, then give

$$h_i(j) = \frac{c_j(c_1 - c_0)}{c_1^2 - c_0 c_2} \quad (0 \le j \le 8), \qquad Q = q_i, \qquad E_0 = I_3(i).$$

In particular, all displayed denominators are nonzero at good starts.

To state the identities as equations on the curve, use the raw-moment polynomials

$$M_0 = 1, \qquad M_{r+1}(k_0, k_1, \dots) = \sum_{t=0}^{r} \binom{r}{t} k_0^{r-t} M_t(k_1, k_2, \dots).$$

Only $M_0, \dots, M_6$ are needed; they involve $k_0, \dots, k_4$. At good starts, $M_r = \mu_i(x^r)$ for $0 \le r \le 6$, by the weight recursion in Section 6. Set

$$H_3 = \det(M_{r+s})_{0\le r,s<3}, \qquad H_4 = \det(M_{r+s})_{0\le r,s<4}, \qquad H_{013} = \det(M_{r+s})_{\substack{r=0,1,2\\ s=0,1,3}}.$$

This H_4 is the polynomial of Lemma 7.3. With $p_3 = \bar{u}_2 - 1 - s_1 - s_2 - s_3$, the coefficient of x^3 in P_4, consider the *normalization* and *resolvent identities*

$$N_0 P_4(-Q) = aH_4, \qquad E_0 P_4(-Q) = a\big((Q - p_3)H_3 - H_{013}\big). \tag{12}$$

Both hold at good starts, where $Q = q_i$, $E_0 = I_3(i)$ and $P_4 = \pi_i$. The first is Lemma 7.3. For the second, write $v_j = h_i(j)$ and $P_4(x) = x^4 + p_3x^3 + p_2x^2 + p_1x + p_0$. Orthogonality of P_4 to $1, x, x^2$ gives $p_0v_j + p_1v_{j+1} + p_2v_{j+2} + p_3v_{j+3} + v_{j+4} = 0$ for $j < 3$, and $q_iv_j + v_{j+1} = \mu_i(x^j) = M_j$. Eliminating v_4, v_5, v_6 with the three relations, a direct expansion gives the polynomial identity

$$\det(v_{r+s})_{r,s<3}\, P_4(-q_i) = (q_i - p_3)\det(M_{r+s})_{r,s<3} - \det(M_{r+s})_{\substack{r=0,1,2\\ s=0,1,3}},$$

and $a\det(v_{r+s})_{r,s<3} = I_3(i)$. Among components containing a good start, we call a component *retained* if both equations, after clearing denominators, hold identically on it.

Lemma 8.2 (Removing components by polynomial equations)**.** *The two identities in* (12)*, after clearing denominators, have degrees at most* 166 *and* 404*, respectively. On a degree-d component, either an identity holds everywhere or it holds at* $166d$ *or* $404d$ *good starts at most, respectively. Consequently each component that is not retained contributes $O(1)$ good starts.*

Proof. Write $Q = Q_{\mathrm{n}}/Q_{\mathrm{d}}$ and $E_0 = E_{\mathrm{n}}/E_{\mathrm{d}}$ using the cofactor expressions above. Each entry of the coefficient matrix has degree at most four in the nodes, so each 8×8 cofactor has degree at most 32, and in particular at most 40, the bound used in the formal proof. Hence

$$\deg Q_{\mathrm{n}}, \deg Q_{\mathrm{d}} \le 40, \qquad \deg E_{\mathrm{n}}, \deg E_{\mathrm{d}} \le 240.$$

The latter bound follows from the 3×3 determinant and the cube in the numerator, and the cube of a quadratic cofactor expression in the denominator. The polynomial $\widehat{P} = Q_{\mathrm{d}}^4 P_4(-Q_{\mathrm{n}}/Q_{\mathrm{d}})$ has degree at most 163: in Newton form its longest product is $Q_{\mathrm{n}} \prod_{j=1}^{3}(Q_{\mathrm{n}} + s_jQ_{\mathrm{d}})$, of degree at most $40 + 3 \cdot 41$, because $s_0 = 0$. The cleared equations are

$$\begin{aligned} \Phi_N &= N_0\widehat{P} - aH_4Q_{\mathrm{d}}^4, \\ \Phi_R &= E_{\mathrm{n}}\widehat{P} - a\big((Q_{\mathrm{n}} - p_3Q_{\mathrm{d}})H_3 - H_{013}Q_{\mathrm{d}}\big)Q_{\mathrm{d}}^3E_{\mathrm{d}}. \end{aligned}$$

Here $\deg H_3 \le 3$, $\deg H_4 \le 6$ and $\deg H_{013} \le 4$. Thus $\deg \Phi_N \le 166$ and $\deg \Phi_R \le 404$.

If a polynomial of degree r does not vanish identically on an integral curve of degree d, its zero set on that curve has at most rd distinct rational points, by the polynomial-section form of Bézout's bound. Every good start satisfies both cleared equations. Moreover, the free nodes and pattern recover $q_i = Q$, and q_i strictly increases along the fixed orbit. Thus distinct good starts of this pattern give distinct free-node triples. Applying the point bound to whichever equation is not an identity proves the claim. The absolute bound $d \le 24$ makes the last estimate uniform. □

8.3. **Directions at infinity.** Call a pattern *exceptional* if two of the five Newton coefficients γ_j sum to zero. Since $\bar{N}_1 > 0$, $\bar{u}_1 > 0$ and $\bar{u}_2 < 0$, every γ_j is nonzero, and a pair sum can vanish only in one of the four cases

$$\bar{N}_1 + \bar{u}_2 = 1, \qquad \bar{N}_1 + \bar{u}_2 = 0, \qquad \bar{u}_1 = 1, \qquad \bar{N}_1 + \bar{u}_1 = 1. \tag{13}$$

Equivalently, the integer pattern coordinates satisfy $N_1 + u_2 = N_0$, $N_1 + u_2 = 0$, $u_1 = N_0$ or $N_1 + u_1 = N_0$.

Lemma 8.3 (Leading nodes and determinant orders)**.** *Let C be an integral component of positive degree of* (11)*. Let P be a closed point above infinity on the normalization of the projective closure of C, write* $\operatorname{ord} = \operatorname{ord}_P$*, and let $e > 0$ be the multiplicity of the hyperplane at infinity at P. For a local parameter ϖ at P, let z_j be the residue of $\varpi^e s_j$, $0 \le j \le 4$. Then the z_j are not all equal and satisfy*

$$\sum_{j=0}^{4} \gamma_j z_j^r = 0 \qquad (0 \le r \le 3).$$

If two of these five leading nodes coincide, the pattern is exceptional. For a nonexceptional pattern, therefore,

$$\begin{aligned} \operatorname{ord}(s_l - s_j) &= -e \quad (0 \le j < l \le 4), \\ \operatorname{ord} \operatorname{Res}(P_4, Z) &= -10e, \\ \operatorname{ord} H_3 = -3e, &\qquad \operatorname{ord} H_4 = -6e. \end{aligned}$$

Positive order denotes a zero and negative order a pole.

Proof. All nonzero rational constants have order zero. The multiplicity e can equivalently be written as

$$e = \max_{1 \le j \le 3} \{ - \operatorname{ord} s_j \} > 0.$$

Thus a polynomial of total node degree at most r has order at least $-re$. The residues z_j lie in the residue field of P; we do not require P to be rational. We have $z_0 = 0$, and at least one of z_1, z_2, z_3 is nonzero.

The leading nodes are distinct. The definitions of the nodes give the moment equations of orders zero and one. The highest-degree part of $f = 0$ gives the equation of order two. Finally, the highest-degree part of $N_0 \operatorname{Res}(P_4, Z) = a\Psi^4$ gives the equation of order three: the resultant has total degree at most ten, whereas the cubic part of Ψ is $\sum_j \gamma_j s_j^3$. Consequently

$$\sum_{j=0}^{4} \gamma_j z_j^r = 0 \qquad (0 \le r \le 3). \tag{14}$$

For the last of these, multiply the fixed-dividend equation by ϖ^{12e} and take residues; its left side vanishes and its right side is $a(\sum_j \gamma_j z_j^3)^4$.

If two nodes coincided, there would be at most four distinct values among the z_j. Grouping the sums in (14) by those values and using the corresponding Vandermonde matrix shows that the sum of the γ_j in each group is zero. Since there are five indices and at least two groups, some group has at most two elements. It cannot have one, because every γ_j is nonzero; it must therefore give a vanishing pair sum. This proves the collision assertion. For a nonexceptional pattern it is impossible, so all five nodes are distinct, and

$$\operatorname{ord}(s_l - s_j) = -e \qquad (0 \le j < l \le 4). \tag{15}$$

The determinant orders. Write $\mathcal{R} = \operatorname{Res}(P_4, Z)$. The leading homogeneous parts needed below are

$$[\mathcal{R}]_{10} = \gamma_0\gamma_1\gamma_2\gamma_3 \prod_{0\le j<l\le 4} (s_l - s_j), \tag{16}$$

$$[H_3]_3 = k_0k_1(k_0 + k_1),$$

$$[H_4]_6 = k_0k_1k_2(k_0 + k_1)(k_1 + k_2)(k_0 + k_1 + k_2). \tag{17}$$

Here $[F]_r$ denotes the homogeneous part of total node degree r. For completeness, the degree-ten formula can be checked without expanding the resultant. In the Newton basis beginning at s_1, s_2, s_3, its 4×4 Bezout matrix for P_4, R_4 has entry degrees at most $4 - \max(i, j)$, for $0 \le i, j < 4$. Only the diagonal term of its determinant can have degree ten. Its four leading diagonal entries are

$$\begin{aligned} &-\gamma_1 s_1(s_1 - s_2)(s_1 - s_3)(s_1 - s_4),\\ &-\gamma_2 s_2(s_2 - s_3)(s_2 - s_4),\\ &-\gamma_3 s_3(s_3 - s_4), \qquad -s_4. \end{aligned}$$

Their product gives the stated Vandermonde; the additional factor x in the second polynomial contributes $P_4(0) = \gamma_0$. The raw-moment recurrence gives

$$H_3 = k_0k_1(k_0 + k_1) + k_0k_2 - k_1^2,$$

and gives the displayed degree-six part of H_4, with remainder of degree at most five. These are finite polynomial identities.

Every factor in these leading products has order $-e$ by (15). The lower-degree terms have strictly larger order, so they cannot cancel the leading terms. Thus

$$\operatorname{ord}\mathcal{R} = -10e, \qquad \operatorname{ord} H_3 = -3e, \qquad \operatorname{ord} H_4 = -6e. \tag{18}$$

□

Corollary 8.4 (Even component degrees)**.** *Every positive-degree nonexceptional component of* (11) *has even degree. At each point above infinity its hyperplane multiplicity e is even, and* $\operatorname{ord}(N_0\Psi) = -5e/2$.

Proof. Taking orders in the fixed-dividend equation and using Lemma 8.3 gives $4 \operatorname{ord}\Psi = -10e$. Since orders are integers and $N_0 \neq 0$, this proves both local assertions. The degree of the curve is the sum of the multiplicities e, weighted by the residue-field degrees, so it is even. The retained identities are not needed here. □

8.4. **Exceptional patterns and line components.** Exceptional patterns may have components of odd degree, including lines, but there are fewer of them.

Lemma 8.5 (Exceptional patterns and lines)**.** *There are $O(X^4 \log(2X)) = O_\eta(a^\eta X^4)$ exceptional patterns. In the range*

$$124928A^4H^5X < a^3, \tag{19}$$

no line component contains a good start.

Proof. Each relation in (13) becomes a vanishing maximal minor after a fixed invertible integer column change. The transformed minor vector still has height $O(X)$. For example, if the original columns are $C_0, \dots, C_4$, then

$$\det(C_1, C_2 + C_0, C_3 - C_0) = p_{123} - p_{013} - p_{012} = N_1 + u_2 - N_0.$$

The other three relations follow by the same column additions and permutations.

A fixed zero minor reduces the Plücker relations to three integer vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ of height $O(X)$ with $\mathbf{x} \times \mathbf{y} = 0$ and $\mathbf{x} \cdot \mathbf{z} = \mathbf{y} \cdot \mathbf{z} = 0$. These vectors are signed rearrangements of the nine remaining minors, so they determine the minor vector. The case $\mathbf{x} = \mathbf{y} = 0$ has $O(X^3)$ choices. Otherwise write $\mathbf{x} = r\mathbf{v}$, $\mathbf{y} = s\mathbf{v}$ with $\mathbf{v}$ primitive. In a dyadic range $R \le |\mathbf{v}|_\infty < 2R$, there are $O(R^3)$ choices for $\mathbf{v}$ and $O((X/R)^2)$ for (r, s). The equation $\mathbf{v} \cdot \mathbf{z} = 0$ has $O(X^2/R)$ solutions in the box. Indeed, take a coordinate of absolute value $n = |\mathbf{v}|_\infty$. Primitivity gives n admissible pairs of the other two coordinates modulo n, and the chosen coordinate is then determined. Since $R \le n \ll X$, this gives $O(X^2/n)$ choices. Each range therefore contributes $O(X^4)$, and summing over $O(\log(2X))$ ranges proves the pattern count. The crude bound $X \le 10^{16} a^7$ absorbs its logarithm into a^η.

Now suppose that a line contained in the zero set of (11) passes through the free-node triple s of a good start, with direction v. Extend v to the five nodes linearly, as in Section 8.1, and write $s(t) = s + tv$. Both equations of (11) hold identically in t.

(1) The resultant has degree at most ten in t, so the t^{12} coefficient of the second equation, $-a\big(\sum_j \gamma_j v_j^3\big)^4$, vanishes. Then $\Psi(s(t))$ has degree at most two, so Ψ^4 has degree at most eight, and the t^{10} coefficient $[\mathcal{R}]_{10}(v)$ of the resultant, given by (16), vanishes too: two coordinates of v coincide.

(2) The direction satisfies the four moment equations of Lemma 8.3: those of orders zero and one by the definition of the nodes, order two from the t^2 coefficient of f, and order three from step (1). The grouping argument in the proof of Lemma 8.3 then shows that the pattern is exceptional, with exactly two groups of equal coordinates, of sizes two and three. After scaling, and in the order of (13), v is one of

$$(0,1,1,0,1), \quad (0,1,1,0,0), \quad (0,0,1,1,0), \quad (0,1,0,0,1).$$

(3) For each of these four directions the resultant has degree at most six in t. This follows from a row and column degree count in the Newton-basis Bezout matrix used in the proof of Lemma 8.3, in which coinciding leading nodes lower the degrees of the corresponding entries. Hence Ψ^4 has degree at most six, $\Psi(s(t))$ is affine in t, and its second difference $\Psi(s+v) + \Psi(s-v) - 2\Psi(s)$ vanishes.

(4) Expanding this second difference, and using $f = 0$ at $s \pm v$, gives respectively

$$-2\bar{N}_1 s_3, \qquad 2(\bar{N}_1 + \bar{u}_1)(k_1 - 1), \qquad 2(\bar{u}_2 - 1)(k_2 - 1), \qquad 2\big((\bar{u}_1 - \bar{u}_2) - (s_4 - s_1)\big).$$

The factors $\bar{N}_1$, $\bar{N}_1 + \bar{u}_1$ and $\bar{u}_2 - 1$ are nonzero, so respectively

$$s_3 = 0, \qquad k_1 = 1, \qquad k_2 = 1, \qquad s_4 - s_1 = \bar{u}_1 - \bar{u}_2.$$

The Lean modules `FixedDividendDirectionExtraction`, `FixedDividendLineResultant` and `FixedDividendLines` contain steps (2)–(4). The first case is impossible at a good start, where $s_3 \ge 3$. The polynomial G in Lemma 7.4 satisfies, by expansion,

$$G \le H^5(12k_1 + 464), \qquad G \le H^5(12k_2 + 464).$$

In the last obstruction, $k_1 + k_2 + k_3 = \bar{u}_1 - \bar{u}_2 = (u_1 - u_2)/N_0 \le 2X$. Thus each surviving case gives $k_1 \le 2X$ or $k_2 \le 2X$. The label bound $N_0 q_i^4 \le aG$, with $N_0 \ge 1$, now gives

$$q_i^4 \le aH^5(24X + 464).$$

But $q_i \ge a/(4A)$ and $24X + 464 \le 488X$ for $X \ge 1$. Condition (19), with $124928 = 4^4 \cdot 488$, gives the strict reverse inequality. □

8.5. **Orders of the reconstructed functions.**

Lemma 8.6 (Orders of the reconstructed functions)**.** *Let C be a retained nonexceptional component of positive degree containing a good start. At every point above infinity, with hyperplane multiplicity e, one has*

$$\operatorname{ord} Q = -\frac{3e}{2}, \qquad \operatorname{ord} E_0 = \frac{3e}{2}.$$

In particular, E_0 is a nonzero rational function vanishing at every point above infinity.

Proof. The good start guarantees that the reconstruction denominators are not identically zero on C. Thus Q and E_0 are rational functions on C, and retention makes both identities (12) identities in its function field.

The quotient order. Write $P_4(x) = x^4 + p_3x^3 + p_2x^2 + p_1x + p_0$. Its Newton expression gives $\deg p_j \le 4 - j$, and hence $\operatorname{ord} p_j \ge -(4-j)e$. The first retained identity gives

$$\operatorname{ord} P_4(-Q) = \operatorname{ord}(aH_4/N_0) = -6e.$$

If $\operatorname{ord} Q \ge -e$, every term of $P_4(-Q)$ would have order at least $-4e$, a contradiction. Put $r = \operatorname{ord} Q < -e$. For $0 \le j < 4$,

$$\operatorname{ord}\big(p_j(-Q)^j\big) - 4r \;\ge\; (4-j)(-e-r) > 0.$$

Thus Q^4 is the unique term of lowest order. It follows that $4r = -6e$, or $\operatorname{ord} Q = -3e/2$.

The reconstructed curvature order. The coefficient $p_3 = \bar{u}_2 - 1 - s_1 - s_2 - s_3$ has order at least $-e$. The raw-moment recurrence also shows that H_{013} has degree at most four: its degree-four part is $(2k_0 + k_1)k_0k_1(k_0 + k_1)$. Consequently

$$\operatorname{ord}(Q - p_3) = -3e/2, \qquad \operatorname{ord}\big((Q-p_3)H_3\big) = -9e/2 < -4e \le \operatorname{ord} H_{013}.$$

There is again a unique term of lowest order. Dividing the second retained identity by $P_4(-Q)$ therefore gives

$$\operatorname{ord} E_0 = -9e/2 - (-6e) = 3e/2.$$

The argument uses the discrete valuation on the normalization, so it also covers branches above singular points of the original curve. □

8.6. **The joint coordinates.** Put $d_j = b_{i+j} - b_{i+j+1}$ and

$$D_{01} = d_0k_1 - d_1k_0, \quad D_{02} = d_0k_2 - d_2k_0, \quad D_{12} = d_1k_2 - d_2k_1.$$

The first of these, D_{01}, is the curvature $I_3(i)$. With $s_j = q_{i+j} - q_i$ and $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$, the Hankel polynomial H_3 of Section 8.2 and its shift are

$$H_3 = k_0r_1 - k_1r_0 = \sigma(k_0, k_1, k_2),$$
$$H_3' = k_1r_2 - k_2r_1 = \sigma(k_1, k_2, k_3),$$

with σ as in Section 5. They are positive at every state at which they are defined. On a retained component, use the reconstructed function E_0 to define

$$E_{12} = \frac{H_3'E_0 - N_0k_1}{H_3}, \qquad E_{02} = \frac{k_0E_{12} + k_2E_0}{k_1}, \qquad W = (E_{12}, -E_{02}, E_0), \qquad J = N_0\Psi.$$

The determinant and cross-product identities give, identically,

$$W \cdot (k_0, k_1, k_2) = 0, \qquad W \cdot (r_0, r_1, r_2) = N_0. \tag{20}$$

At a good start these rational functions take the integer values $W = (D_{12}, -D_{02}, D_{01})$. Changing the rank-five Gram basis to $1, x, x^2, x^3, P_4$ gives

$$\det\big(\nu_i(x^{r+s})\big)_{r,s<5} = \det\big(\nu_i(x^{r+s})\big)_{r,s<4}\, \nu_i(P_4^2).$$

Consequently $J = I_5(i)$ is a positive integer at good starts. The heights of these integers are controlled by

$$S = \left\lfloor \frac{4352A^3H^3}{a^2} \right\rfloor, \qquad Y = \left\lfloor \frac{1259991040A^5H^{10}}{a^4} \right\rfloor.$$

Lemma 8.7 (Heights of the joint coordinates)**.** *At good starts, $|W_j| \le S$ and $1 \le J \le Y$.*

Proof. Write $q = q_i$, and recall $q \ge a/(4A)$ (proof of Lemma 7.4). For $j \le 3$, apply μ_{i+j} to the identity $(q+y)^{-1} = q^{-1} - yq^{-2} + y^2/(q^2(q+y))$ with $y = s_j + x$, as in Lemma 7.7. Since $\mu_{i+j}((s_j+x)^2) = s_j^2 + 2s_j + k_j + 1 \le 17H^2$, this gives

$$\frac{b_{i+j}}{a} = \frac{1}{q} - \frac{s_j+1}{q^2} + \epsilon_j, \qquad 0 \le \epsilon_j \le 17H^2/q^3.$$

Subtracting consecutive expansions, $d_j/a = k_j/q^2 + \delta_j$ with $|\delta_j| \le 34H^2/q^3$. In $D_{jl} = d_jk_l - d_lk_j$ the leading terms cancel, so $|D_{jl}|q^3 \le 68aH^3$, and $q \ge a/(4A)$ gives $|D_{jl}| \le S$, including the floor because the minors are integers.

For $J = I_5(i)$, argue as in the proof of Lemma 7.4 with five Newton polynomials instead of four: Hadamard's inequality and $\nu_i \le \mu_i/q$ give $Jq^5 \le a\prod_{j<5}\mu_i(F_j^2)$. By the shift rule, $\mu_i(F_j^2) = \mu_{i+j}\big(F_j(x+s_j)\big)$ is a polynomial of degree j in the gaps, and the gap bound gives $\mu_i(F_j^2) \le c_jH^j$ with $(c_0,\dots,c_4) = (1,2,10,77,799)$. Hence $Jq^5 \le 1230460\,aH^{10}$, and $q \ge a/(4A)$ gives $J \le Y$. □

Corollary 8.8 (Orders of the joint coordinates)**.** *In the setting of Lemma 8.6, the joint coordinates satisfy*

$$\operatorname{ord} W_0 = \operatorname{ord} W_2 = 3e/2, \qquad \operatorname{ord} W_1 \ge 3e/2, \qquad \operatorname{ord} J = -5e/2. \tag{21}$$

Proof. The cubic leading part of H_3' is $k_1k_2(k_1+k_2)$. By (15), it has order $-3e$, whereas the quadratic remainder $k_1k_3 - k_2^2$ has order at least $-2e$. Thus $\operatorname{ord} H_3 = \operatorname{ord} H_3' = -3e$ and $\operatorname{ord} k_1 = -e$. In the numerator defining E_{12} the two terms have different orders:

$$\operatorname{ord}(H_3'E_0) = -3e/2 < -e = \operatorname{ord}(N_0k_1).$$

Therefore $\operatorname{ord} E_{12} = 3e/2$. Each term in $k_0E_{12} + k_2E_0$ has order $e/2$, so their sum has order at least $e/2$. Dividing by k_1 gives $\operatorname{ord} E_{02} \ge 3e/2$. Cancellation here is harmless: equality for the first and third coordinates already gives $\min_j \operatorname{ord} W_j = 3e/2$. The formula for $J = N_0\Psi$ is Corollary 8.4. □

8.7. **Counting the nonexceptional conics.**

Proposition 8.9 (Conics)**.** *A nonexceptional component of degree two contains the free-node triples of $O_\eta(a^\eta)$ good starts, with an implied constant independent of the pattern and the component. Consequently all nonexceptional conics together contribute $O_\eta(a^\eta X^5)$ good starts.*

The idea is short. A component that is not retained contributes $O(1)$ good starts (Lemma 8.2). A retained conic through three free-node triples of good starts lies in a plane, and Corollary 8.4 leaves it a single rational branch at infinity, of multiplicity two. So it is a parabola, the image of a quadratic polynomial map $t \mapsto (\xi_0(t), \xi_1(t), \xi_2(t))/\Delta_c$ whose parameter t is an integer at every integer point. By Lemma 8.6, the reconstructed curvature E_0 is a rational function of t with a zero of order three at infinity, and it takes the integer value $I_3(i)$ at each good start. Its nonconstant denominator must then divide a fixed nonzero resultant at each such t, which leaves $a^{o(1)}$ possibilities. Appendix A gives the details, including the uniform bounds on the coefficients.

8.8. **The extra degree and the powered graph.** The conic count leaves retained nonexceptional components of degree at least four. Fix such a component C of degree d. The additional integer coordinates will increase its degree without increasing their heights too much. The key is one extra zero over the affine part of C.

All divisors in this argument are on the normalization $\widetilde{C}$ of the projective closure of C, over $\mathbb{Q}$. For a closed point P of $\widetilde{C}$, write $\deg P = [\kappa(P) : \mathbb{Q}]$, where $\kappa(P)$ is the residue field; thus $\deg(\sum_P a_P[P]) = \sum_P a_P \deg P$. Let $E_\infty = \sum_{P|\infty} e_P[P]$ be the pullback of the original hyperplane at infinity. Then $\sum_{P|\infty} e_P \deg P = d$. We do not require the closed points, or the branches over a rational point of C, to be rational.

Let $\mathcal{F}$ be the pole envelope of the curvature coordinates:

$$\mathcal{F} = \sum_P \max\{0, -\operatorname{ord}_P(W_0), -\operatorname{ord}_P(W_1), -\operatorname{ord}_P(W_2)\}[P],$$

with $\operatorname{ord}_P(0) = +\infty$. The orders in (21) show that $\mathcal{F}$ is supported over the affine part of C, and hence is disjoint from E_∞.

Lemma 8.10 (An additional affine zero)**.** *There is a rational linear combination ϕ of W_0, W_1, W_2 which is not the zero function and has a positive-order zero at a closed point P_0 of $\widetilde{C}$ outside E_∞.*

Proof. First, the W_j cannot all be scalar multiples, in $\mathbb{Q}(C)$, of a fixed nonzero vector $v \in \mathbb{Q}^3$. Such a relation, together with $W \cdot (k_0, k_1, k_2) = 0$, would put C in the plane $v_0k_0+v_1k_1+v_2k_2 = 0$. This is a genuine plane: the first three gaps are an invertible linear change of the three node coordinates. The pattern quadratic is irreducible, so its restriction to this plane is nonzero and has degree at most two. Every curve contained in this plane section therefore has degree at most two, contrary to $d \geq 4$.

Choose the free-node triple $z_* \in C(\mathbb{Q})$ of a good start. All the W_j are regular there, because their common reconstruction denominator does not vanish at a good start. Write $w_j = W_j(z_*)$. The identity $W \cdot (r_0, r_1, r_2) = N_0 \neq 0$ gives $w \neq 0$. Choose h with $w_h \neq 0$. At least one function $w_hW_j - w_jW_h$ is nonzero: otherwise $W = (W_h/w_h)w$, contradicting the preceding paragraph. Take such a function for ϕ. It has rational coefficients and satisfies $\phi(z_*) = 0$.

This zero survives normalization even if z_* is singular. Indeed, write $\phi = g/g_0$ near z_*, with g not identically zero on C, $g(z_*) = 0$ and $g_0(z_*) \neq 0$. The finite normalization has a closed point P_0 above z_*. In its discrete valuation ring the numerator is in the maximal ideal and the denominator is a unit, so $\operatorname{ord}_{P_0}(\phi) > 0$. Since z_* is affine, P_0 does not lie in E_∞. □

Lemma 8.11 (The curvature pole degree)**.** *The pole envelope satisfies*

$$\deg \mathcal{F} \geq \frac{3d}{2} + 1. \tag{22}$$

Proof. Use the function ϕ from Lemma 8.10. At every point over infinity, each W_j has order at least $3e_P/2$. The same is true of a rational linear combination; cancellation can only increase its zero order. Consequently

$$(\phi)_0 \;\geq\; \frac{3}{2}E_\infty + [P_0], \qquad (\phi)_\infty \;\leq\; \mathcal{F}.$$

The first divisor is integral because every e_P is even. The function ϕ is nonzero and has a zero, so its principal divisor has equal zero and pole degrees. Hence

$$\deg \mathcal{F} \;\geq\; \deg(\phi)_\infty = \deg(\phi)_0 \;\geq\; \frac{3}{2}\sum_{P|\infty} e_P \deg P + \deg P_0 \;\geq\; \frac{3d}{2} + 1.$$

The last inequality uses only $\deg P_0 \geq 1$, not that P_0 is rational. □

For positive integers m, n, let $\Gamma_{m,n} \subseteq \mathbb{A}^7$ be the closure of the graph obtained by adjoining W_0^m, W_1^m, W_2^m, J^n to the three original node coordinates. The graph is taken on the common domain of these rational functions. It is an integral curve, and projection to the first three coordinates is birational: on that domain the four remaining coordinates are uniquely determined. Thus $\Gamma_{m,n}$ and C have the same function field. Taking powers does not lose this property, because the original coordinates remain in the graph.

Lemma 8.12 (Degree of the powered graph)**.** *If $\delta = \deg \Gamma_{m,n}$, then*

$$2\delta \geq (5n+3m)d + 2m, \qquad \delta \geq 10n + 7m =: \delta_{\min}. \tag{23}$$

Moreover,

$$\delta \leq \big(247(m+n)+1\big)d.$$

For $m = n = 1$ one has the sharper upper bound $\delta \leq 247d$ and the lower bound $\delta \geq 4d+1 \geq 17$.

Proof. Lower bound. At a point P over the old infinity hyperplane, the original node coordinates have pole orders at most e_P, while J^n has pole order $5ne_P/2$ and all W_j^m vanish. Since $n \geq 1$, the pole envelope of the seven graph coordinates at P is exactly $5ne_P/2$. Over the affine part, the node coordinates and J are regular: $J = N_0\Psi$ is polynomial in the nodes. There the envelope is exactly m times the curvature envelope. The complete envelope of the graph coordinates is therefore

$$\mathcal{F}_{m,n} = \frac{5n}{2}E_\infty + m\mathcal{F}.$$

There is a single rational linear combination λ of the seven coordinates whose pole divisor is $\mathcal{F}_{m,n}$. To see this, fix a point P in the finite support of $\mathcal{F}_{m,n}$ and multiply all seven coordinates by a local uniformizer to the power $\mathcal{F}_{m,n}(P)$. Their residues in $\kappa(P)$ are not all zero. Cancellation of the leading pole is therefore a proper $\mathbb{Q}$-linear condition on the seven coefficients. The finitely many such proper subspaces do not cover $\mathbb{Q}^7$. Choosing coefficients outside their union gives the asserted simultaneous absence of cancellation, also when $\kappa(P) \neq \mathbb{Q}$.

As a linear function on $\Gamma_{m,n}$, λ has degree at most δ. For example, homogenize it on the projective closure of the graph: its pencil with the homogenizing coordinate consists of hyperplane sections of degree δ. Removing common zeros can only decrease the degree of the resulting map to $\mathbb{P}^1$. Birational projection identifies its function field with $\mathbb{Q}(C)$, so this degree is $\deg(\lambda)_\infty = \deg \mathcal{F}_{m,n}$. It follows that

$$\delta \;\geq\; \frac{5n}{2}d + m\deg\mathcal{F} \;\geq\; \frac{5n+3m}{2}d + m.$$

This is the first inequality in (23); $d \geq 4$ gives the second. The auxiliary linear combination is used only to measure degree: no bound on its coefficients enters the integer-point box.

Upper bound. This is a routine count of polynomial degrees, given in Appendix B. □

Every good start whose free-node triple lies on C gives an integer point on this graph. By Lemma 8.7, its coordinates have absolute values at most

$$\max(3H, S^m, Y^n).$$

The original node coordinates also show that distinct such node triples remain distinct after adjoining the powers.

8.9. **The window bound and the final exponents.** We use the coefficient-uniform affine-curve theorem of Castryck–Cluckers–Dittmann–Nguyen [3, Theorem 3], also stated as [4, Corollary 2.2]: an integral affine curve C over $\mathbb{Q}$ of degree δ in $\mathbb{A}^r$, with $r \geq 2$ fixed, has

$$N_{\mathrm{aff}}(C, B) \ll_r \delta^3 B^{1/\delta}(\log B + \delta), \qquad B \geq 1, \tag{24}$$

integer points in the box $[-B, B]^r$, with an implied constant independent of the coefficients of C. Its formal proof is discussed in Section 12.

Proposition 8.13 (Window bound)**.** *Fix positive integers m, n and $\eta > 0$, and put $\mathcal{B} = \max(H, S^m, Y^n)$. For a window run in $[A, 2A)$ with $T \geq 36$ satisfying* (19)*,*

$$T \ll_{m,n,\eta} a^{\eta}\left(X^5 + X^4 H^{1/2} + X^5 \mathcal{B}^{1/\delta_{\min}}\right). \tag{25}$$

Proof. At least $T/2$ starts are good, and for each pattern the free-node triples of its good starts are distinct (Lemma 8.2). By Lemma 8.1, these triples lie on at most twenty-four integral curves of degree at most twenty-four, and we bound the good starts on each curve, allowing overlaps. All the heights H, S, Y and X are at most a fixed power of a, so the logarithm in (24) is $O_\eta(a^\eta)$.

- *Exceptional patterns.* There are $O_\eta(a^\eta X^4)$ of them (Lemma 8.5). Under (19) no line contains a good start, so every relevant curve has degree $2 \leq \delta \leq 24$, and (24) in $\mathbb{A}^3$ with $B = 3H$ gives $O_\eta(a^\eta H^{1/2})$ good starts per pattern. This gives the term $X^4 H^{1/2}$.
- *Nonexceptional patterns: components that are not retained, and conics.* By Lemma 8.2 and Section 8.7, these contribute $O_\eta(a^\eta)$ per pattern, hence $O_\eta(a^\eta X^5)$ in all.
- *Remaining nonexceptional components.* Lines do not occur, by Lemma 8.5, and odd degrees do not occur, by Corollary 8.4, so these components have degree $d \geq 4$. Their good starts give distinct integer points of $\Gamma_{m,n} \subseteq \mathbb{A}^7$ in a box of side $\max(3H, S^m, Y^n) \leq 3\mathcal{B}$. By Lemma 8.12, the degree δ of $\Gamma_{m,n}$ satisfies $\delta_{\min} \leq \delta \leq 24(247(m+n)+1)$, so (24) gives $O_{m,n,\eta}(a^\eta \mathcal{B}^{1/\delta_{\min}})$ good starts per component. This gives the term $X^5 \mathcal{B}^{1/\delta_{\min}}$.

The added coordinates W_j and J are functions of the free nodes on each component. They therefore introduce no further freely counted coordinates, and the pattern count is not enlarged. □

Corollary 8.14. *For fixed positive integers m, n and every $\varepsilon > 0$,*

$$P(a, b) \ll_{m,n,\varepsilon} a^{E(m,n)+\varepsilon}, \qquad E(m, n) = \max\left(\frac{15\delta_{\min} + m}{51\delta_{\min} + 3m},\ \frac{15\delta_{\min} + 6n}{51\delta_{\min} + 20n}\right),$$

where $\delta_{\min} = 10n + 7m$. In particular $E(1, 1) = 128/435$ and $E(50, 51) = 2201/7480$.

Proof. We bound the number T of terms in each dyadic window, eliminating the window scale A. If $T < 36$ there is nothing to prove. If $AT > 28a$, then Proposition 5.3 gives $T^4 \ll a$. Otherwise $HAT \leq 56a$, and Proposition 4.2 gives $T^2 \leq 8\Delta A$. Substituting $H \leq 56a/(AT)$ and then $A \geq T^2/(8\Delta)$ into the definitions of X, S and Y gives

$$H \ll \frac{\Delta a}{T^3}, \qquad X \ll \frac{\Delta^2 a^3}{T^{10}}, \qquad S \ll \frac{a}{T^3}, \qquad Y \ll \frac{\Delta^5 a^6}{T^{20}}. \tag{26}$$

Every power of Δ is $O(a^\eta)$, and we absorb such factors into $a^{O(\eta)}$.

- If (19) fails, then $124928 A^4 H^5 X \geq a^3$. With $Xa^3 \leq 3609600 A^4 H^6$ this gives $A^8 H^{11} \gg a^6$, and $H \leq 56a/(AT)$ turns it into $A^3 T^{11} \ll a^5$, so $T^{17} \ll \Delta^3 a^5$.
- Otherwise Proposition 8.13 applies. By (26), its first term gives $T \ll a^{O(\eta)} a^{15}/T^{50}$, so $T^{51} \ll a^{15+O(\eta)}$. Its second term gives $T \ll a^{O(\eta)} a^{12+1/2}/T^{40+3/2}$, so $T^{85} \ll a^{25+O(\eta)}$.

- For the graph term, $T \ll a^{O(\eta)} X^5 \mathcal{B}^{1/\delta_{\min}}$ gives $T^{51\delta_{\min}} \ll a^{O(\eta)} a^{15\delta_{\min}} \mathcal{B}$. Since $H \geq 1$, $\max(H, S^m) \leq \max(H, S)^m \ll a^{O(\eta)} (a/T^3)^m$, so if $\mathcal{B} = \max(H, S^m)$ then $T^{51\delta_{\min}+3m} \ll a^{15\delta_{\min}+m+O(\eta)}$. If $\mathcal{B} = Y^n$ then $T^{51\delta_{\min}+20n} \ll a^{15\delta_{\min}+6n+O(\eta)}$.

The first three exponents all equal $5/17$, both fractions in $E(m,n)$ exceed $5/17$, and $T^4 \ll a$ is stronger still. Choosing η small in terms of ε, m and n, every window has $T \ll a^{E(m,n)+\varepsilon/2}$, and summing over the $O(\log a)$ windows proves the bound. For $m = n = 1$ the two fractions are $256/870 = 128/435$ and $261/887$, and the first is larger. For $(m,n) = (50, 51)$ the larger fraction is $2201/7480$. □

Proof of Theorem 1.3. Set $r = n/m$. The two fractions in $E(m,n)$ become

$$E_S(r) = \frac{150r + 106}{510r + 360}, \qquad E_Y(r) = \frac{156r + 105}{530r + 357}.$$

The first decreases and the second increases on $r > 0$. Their crossing satisfies $30r^2 - 10r - 21 = 0$, so $r_* = (5+\sqrt{655})/30$ and their common value is $\alpha_* = (1310-\sqrt{655})/4365$. Positive rational ratios approximate r_*; choose the fixed positive integers m, n to make $E(m,n) < \alpha_* + \varepsilon/2$, and apply Corollary 8.14 with loss $\varepsilon/2$. The powers and all constants depend only on ε, not on a, b. □

Remark 8.15. In terms of the scales, at the critical point $T \approx a^{\alpha_*}$ one has $A \approx a^{2\alpha_*}$, $H \approx S \approx a^{1-3\alpha_*} \approx a^{0.117}$, $X \approx a^{3-10\alpha_*} \approx a^{0.057}$ and $Y \approx a^{6-20\alpha_*} \approx a^{0.115}$. The pattern count $X^5 \approx a^{0.287}$ alone would give the exponent $5/17$. The graph term multiplies it by $\mathcal{B}^{1/\delta_{\min}}$, and at $n/m \approx r_*$ the two box radii S^m and Y^n balance, at $\mathcal{B} \approx a^{0.117m}$. This gives the extra factor $a^{0.117/(10r_*+7)} \approx a^{0.0068}$.

9. Exterior lattice coordinates

We now prove Theorem 1.2. We keep the notation and the conventions on constants of Section 8, and write $h = \delta_{\min} = 10n + 7m$ throughout. Only the count of good starts on retained nonexceptional components of degree at least four changes. Proposition 9.11 replaces the term $X^5 \mathcal{B}^{1/h}$ of Proposition 8.13 by

$$X^{5-1/(3h)} H^{1/h} + X^{5-2m/(3h)} S'^{m/h} + X^5 Y^{n/h},$$

where S' differs from S only in its constant (Lemma 9.8). In effect H is replaced by $H/X^{1/3}$ and S by $S/X^{2/3}$, while Y is unchanged. The node term stays below the critical exponent, so of the two fractions balanced in the proof of Theorem 1.3 only the curvature fraction improves, and the crossing moves from $\alpha_* = 0.2942513\ldots$ to $\gamma_* = 0.2942294\ldots$. Every pattern has $\lambda(p) \ll X^{1/3}$ (Lemma 9.2), and the weighted count of Lemma 9.6 shows that this saving is available on average: the patterns with an unusually short vector are few enough to pay for their larger boxes.

Section 9.1 attaches to each pattern p a rank-three lattice $\Lambda_p \subseteq \mathbb{Z}^3$ that contains the restrictions of the rows of $\widehat{\mathcal{M}}_i$ to three pivot columns, and gives it a reduced basis. Section 9.2 shows that few patterns have a short vector in Λ_p. Section 9.3 writes the nodes and the curvature coordinates in the reduced basis. Section 9.4 checks that this linear change of coordinates keeps the degree bounds of Lemma 8.12, and Section 9.5 sums over patterns and optimizes the exponent.

9.1. **The lattice of a pattern.** Let $p \neq 0$ be the minor vector of a 3×5 integer matrix $\mathcal{M}$, with pivot π as in the proof of Proposition 7.9; so $|p_\pi|$ is maximal among the coordinates of p. Put $\kappa = |p_\pi| \geq 1$, let $j_1 < j_2 < j_3$ be the pivot columns and $j_4 < j_5$ the other two, and call $x_\pi = (x_{j_1}, x_{j_2}, x_{j_3})$ the *pivot part* of a row x of $\mathcal{M}$. Let $u, v \in \mathbb{Z}^3$ be as in that proof: u_r (v_r) is the minor with the r-th pivot column replaced by column j_4 (j_5), with the sign of the column permutation. So the coordinates of u and v are, up to sign, coordinates of p, and $p_\pi x_{j_4} = u \cdot x_\pi$ and $p_\pi x_{j_5} = v \cdot x_\pi$ for every row x (Lemma 9.1). Let $\Lambda_p \subseteq \mathbb{Z}^3$ be the subgroup of x with $\kappa \mid u \cdot x$ and $\kappa \mid v \cdot x$. It contains $\kappa\mathbb{Z}^3$. Write $|x|$ for the Euclidean norm and $|x|_\infty$ for the sup norm, and let the *minimum* $\lambda(p)$ be the least sup norm of a nonzero vector of Λ_p. Then $|x| \geq \lambda(p)$ for every nonzero $x \in \Lambda_p$.

Lemma 9.1 (Lifting)**.** *For a column l, let $L_l \in \mathbb{Z}^3$ be $p_\pi \mathbf{e}_r$ if $l = j_r$ is a pivot column, u if $l = j_4$ and v if $l = j_5$, where $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is the standard basis. For every row x of $\mathcal{M}$ and every column l, $p_\pi x_l = L_l \cdot x_\pi$. Moreover $\det(L_0, L_1, L_2) = p_\pi^2 p_{012}$, and $|L_l|_\infty \leq \kappa$.*

Proof. The rows L_l are the rows of $\mathcal{M}^\mathsf{T} \operatorname{adj}(\mathcal{M}_\pi^\mathsf{T})$, where $\mathcal{M}_\pi$ is the pivot 3×3 block. This is Cramer's rule. Both identities are polynomial identities in the fifteen entries, checked symbolically for each of the ten pivots. The bound holds because the coordinates of u and v are, up to sign, coordinates of p, and $|p_\pi|$ is maximal. □

So the pivot parts x_π of the rows of $\mathcal{M}$ lie in Λ_p.

Lemma 9.2 (Index)**.** *If K is a nonnegative integer and $(K+1)^3 > \kappa$, then $\lambda(p) \leq K$.*

Proof. Write $g = \gcd(\kappa, u)$, $\ell = \kappa/g$ and $u = gu_0$. The proof of Proposition 7.9 gives $v = \mu u_0 + \ell y$ with $\mu \in \mathbb{Z}$ and $y \in \mathbb{Z}^3$. Put $\psi(x) = \big(u_0 \cdot x \bmod \ell,\ (\mu \lfloor u_0 \cdot x/\ell \rfloor + y \cdot x) \bmod g\big)$, which takes at most $\ell g = \kappa$ values. The box $[0, K]^3$ has $(K+1)^3 > \kappa$ points, so by pigeonhole there are points $x \neq x'$ of $[0, K]^3$ with $\psi(x) = \psi(x')$. Then $u_0 \cdot (x - x') = \ell(f - f')$, where f, f' are the two floors, so $u \cdot (x - x') = \kappa(f - f')$, and $v \cdot (x - x') = \ell\big(\mu(f - f') + y \cdot (x - x')\big)$ with the bracket divisible by g. Hence $x - x'$ is a nonzero vector of Λ_p with sup norm at most K. □

The next lemma gives a reduced basis in dimension three. Its absolute constant will not affect the exponents; the coordinate bound is what we use.

Lemma 9.3 (Reduction)**.** *Let $\Lambda \subseteq \mathbb{Z}^3$ be a subgroup containing $\kappa\mathbb{Z}^3$, $\kappa \geq 1$. There is a basis $\beta_1, \beta_2, \beta_3$ of Λ with $V = \det(\beta_1, \beta_2, \beta_3)$, $0 < V \leq \kappa^3$ and*

$$27\,|\beta_1|^2 |\beta_2|^2 |\beta_3|^2 \leq 64\,V^2.$$

Hence every $x \in \Lambda$ is $\sum \xi_i \beta_i$ with $\xi \in \mathbb{Z}^3$ and $27|\beta_i|^2 \xi_i^2 \leq 64|x|^2$.

The proof is given in Appendix C.

9.2. **Few patterns have a short vector.**

Lemma 9.4 (Divisibility count)**.** *For $\ell \geq 1$ and $x \in \mathbb{Z}^3$, at most $27\ell^2 \gcd(\ell, x)$ vectors $y \in [-\ell, \ell]^3$ have $\ell \mid y \cdot x$. Here $\gcd(\ell, x) = \gcd(\ell, x_1, x_2, x_3)$.*

Proof. Each residue class modulo ℓ meets $[-\ell, \ell]^3$ in at most 3^3 points, so it suffices to show that the set $\mathcal{K}$ of solutions in $[0, \ell)^3$ has at most $\ell^2 e$ elements, where $e = \gcd(\ell, x)$. By Bézout choose $z \in \mathbb{Z}^3$ with $z \cdot x \equiv e \pmod{\ell}$. The map $(y, j) \mapsto (y + jz) \bmod \ell$ is injective on $\mathcal{K} \times [0, \ell/e)$: equal images give $\ell \mid (j - j')e$, so $j = j'$, and then $y = y'$. Hence $|\mathcal{K}|\,\ell/e \leq \ell^3$. □

Proposition 9.5 (Short vectors)**.** *Let $\mathcal{S}$ be a set of nonzero minor vectors with coordinates in $[-X, X]$, and let $\tau(j) \leq \tau_0$ for $1 \leq j \leq X$. Then for every $\rho \geq 1$,*

$$\#\{p \in \mathcal{S} : \lambda(p) \leq \rho\} \leq 393660\,\rho^3 X^4 \tau_0^2.$$

Proof. If $\lambda(p) \le \rho$, some nonzero $x \in [-\rho,\rho]^3$ lies in Λ_p, so $\kappa \mid u\cdot x$, where $\kappa = |p_\pi| \le X$; we drop the condition $\kappa \mid v\cdot x$. For fixed x, the encoding $p \mapsto (\pi, \operatorname{sgn} p_\pi, \kappa, u, v)$ of the proof of Proposition 7.9 gives at most $20\sum_{\kappa\le X} N(\kappa,x)$ such p, where $N(\kappa,x)$ counts pairs (u,v) in $[-\kappa,\kappa]^3$ with $\kappa \mid u\times v$ and $\kappa \mid u\cdot x$. By the proof of Proposition 7.9, for each u there are at most $27\kappa\gcd(\kappa,u)^2$ vectors v. Group the u by $g=\gcd(\kappa,u)$ and write $u = gu_0$ with $(\kappa/g) \mid u_0\cdot x$. Lemma 9.4 with $\ell = \kappa/g$, and $\gcd(\kappa/g,x) \le \gcd(\kappa,x)$, give

$$\sum_u g^2 \le \sum_{g\mid\kappa} g^2 \cdot 27\Big(\frac{\kappa}{g}\Big)^2 \gcd(\kappa,x) = 27\kappa^2\tau(\kappa)\gcd(\kappa,x),$$

so $N(\kappa,x) \le 729\kappa^3\tau(\kappa)\gcd(\kappa,x)$. Next, $\gcd(\kappa,x) = \sum_{e\mid\gcd(\kappa,x)}\varphi(e) \le \sum_{e\mid\kappa,\, e\mid x} e$, and swapping the sums,

$$\sum_{0<|x|_\infty\le\rho} \gcd(\kappa,x) \le \sum_{e\mid\kappa} e\,\#\{x\neq 0 : |x|_\infty\le\rho,\ e\mid x\} \le 27\rho^3\tau(\kappa).$$

Indeed, a divisor $e>\rho$ of κ divides no nonzero x in the box, and for $e\le\rho$ the multiples of e number at most $(2\rho/e+1)^3$, with $e(2\rho/e+1)^3 \le 27\rho^3$. Sum over x and then over $\kappa\le X$, using $\sum_{\kappa\le X}\kappa^3 \le X^4$. The product of constants is $20\cdot 729\cdot 27 = 393660$. □

Lemma 9.6 (Fractional weights)**.** *Let $\mathcal{S}$ be a set of nonzero minor vectors with coordinates in $[-X,X]$, $X\ge 1$, and let $\tau_0\ge 1$ satisfy $\tau(j)\le\tau_0$ for $1\le j\le X$. Then for $0\le t\le 1$,*

$$\sum_{p\in\mathcal{S}} \lambda(p)^{-t} \le 3163861\,\tau_0^2 X^{5-t/3}.$$

Proof. Write $C_0 = 393660$ for the constant of Proposition 9.5 and $C_1 = 14581 + 8C_0 = 3163861$. Choose an integer K with $X \le K^3 \le 8X$. By Lemma 9.2, $1\le\lambda(p)\le K$ for $p\in\mathcal{S}$. Writing $1/\lambda = 1/K + \sum_{\lambda\le j<K} 1/(j(j+1))$ and using Proposition 9.5 with $\rho = j$,

$$K\sum_{p\in\mathcal{S}}\frac{1}{\lambda(p)} \le \#\mathcal{S} + C_0 K X^4\tau_0^2 \sum_{1\le j<K}\frac{j^2}{j+1} \le \#\mathcal{S} + C_0 X^4\tau_0^2 K^3.$$

The proof of Proposition 7.9 gives $\#\mathcal{S} \le 14581\,\tau_0 X^5$. Since $K^3\le 8X$ and $\tau_0\ge 1$, the right side is at most $C_1\tau_0^2X^5$. Dividing by $K\ge X^{1/3}$ proves the case $t=1$. For $0<t<1$, Hölder's inequality with exponents $1/(1-t)$ and $1/t$ gives $\sum\lambda^{-t} \le (\#\mathcal{S})^{1-t}\big(\sum\lambda^{-1}\big)^t$, which is the claim; the cases $t=0$ and $t=1$ are the two bounds above. □

9.3. **Exterior coordinates.**

Lemma 9.7 (Exterior coordinates)**.** *Let β_1,β_2,β_3 be a basis of $\Lambda\subseteq\mathbb{Z}^3$ as in Lemma 9.3, with $|\beta_i|\ge\lambda>0$. Let $x,x'\in\Lambda$ have coordinates $\xi,\xi'\in\mathbb{Z}^3$, and put $w = x\times x'$ and $c = \xi\times\xi'\in\mathbb{Z}^3$. Then*

$$w = c_1\,\beta_2\times\beta_3 + c_2\,\beta_3\times\beta_1 + c_3\,\beta_1\times\beta_2, \qquad \beta_i\cdot w = Vc_i, \qquad 27\lambda^4 c_i^2 \le 64|w|^2.$$

Proof. The first identity follows by expanding $(\sum\xi_i\beta_i)\times(\sum\xi'_i\beta_i)$ bilinearly, and taking the dot product with β_i gives the second. By Cauchy–Schwarz and Lemma 9.3,

$$27|\beta_1|^2|\beta_2|^2|\beta_3|^2c_i^2 \le 64V^2c_i^2 = 64(\beta_i\cdot w)^2 \le 64|\beta_i|^2|w|^2.$$

Cancelling $|\beta_i|^2$ and using $|\beta_j|\ge\lambda$ for the other two vectors gives the claim. □

The lemma replaces the bounds $|\xi| \ll |x|/\lambda$ and $|\xi'| \ll |x'|/\lambda$, which would give $|c| \ll |x||x'|/\lambda^2$, by $|c| \ll |x \times x'|/\lambda^2$. This matters because the drop row is large while the curvature minors are small.

The pivot columns of a pattern may include columns 3 and 4, so the cross product of the pivot parts involves minors D_{jl} with $l \geq 3$. We therefore bound all ten minors, not only the three of Lemma 8.7.

Lemma 9.8 (All curvature minors)**.** *At a good start, all ten minors $D_{jl} = d_j k_l - d_l k_j$ with $0 \leq j < l \leq 4$ satisfy*

$$|D_{jl}| \leq S' := \left\lfloor \frac{9472 A^3 H^3}{a^2} \right\rfloor.$$

Proof. Argue as in Lemma 8.7, using the expansions for $j \leq 5$, hence d_j for $j \leq 4$. The gaps $k_0, \dots, k_5$ are at most H, so $s_j \leq 5H$ and $\mu_{i+j}((s_j+x)^2) = s_j^2 + 2s_j + k_j + 1 \leq (5^2 + 2\cdot 5 + 2)H^2 = 37H^2$. This gives $d_j/a = k_j/q^2 + \delta_j$ with $|\delta_j| \leq 2\cdot 37 H^2/q^3$ for $j \leq 4$, hence $|D_{jl}|q^3 \leq 4\cdot 37\, aH^3 = 148\, aH^3$, and $q \geq a/(4A)$ gives $|D_{jl}| \leq 4^3 \cdot 148 A^3 H^3/a^2 = 9472 A^3 H^3/a^2$. □

Now fix a pattern p that occurs at a good start, and let $\beta_1, \beta_2, \beta_3$ be a basis of Λ_p as in Lemma 9.3. Write $\lambda = \lambda(p)$. At a good start i with pattern p, the pivot parts d_π and k_π of the drop and gap rows of $\widehat{\mathcal{M}}_i$ lie in Λ_p, with coordinates ξ and ζ. Put

$$c = \xi \times \zeta, \qquad w = d_\pi \times k_\pi.$$

The entries of w are, up to sign, the minors D_{jl} with j, l pivot columns. The next lemma shows that $c = G_p W$ for a constant matrix G_p. We use c rather than $G_p W$ because c is visibly integral, while G_p has rational entries.

Lemma 9.9 (Coordinates in the reduced basis)**.** *There are matrices $B_p, G_p \in \mathrm{GL}_3(\mathbb{Q})$, depending only on p, such that at every good start with pattern p*

$$(s_1, s_2, s_3) = B_p \zeta, \qquad c = G_p W, \qquad |\zeta|_\infty \leq \frac{3H}{\lambda}, \qquad |c|_\infty \leq \frac{8S'}{3\lambda^2}.$$

Proof. By Lemma 9.1, $p_\pi(k_0, k_1, k_2) = \mathcal{L} k_\pi$, where $\mathcal{L}$ has rows L_0, L_1, L_2 and $\det \mathcal{L} = p_\pi^2 p_{012} \neq 0$, since $p_{012} = N_0 = I_4(i) \geq 1$ at a good start. With $k_\pi = (\beta_1\, \beta_2\, \beta_3)\zeta$ and the cumulative sums $s_j = k_0 + \cdots + k_{j-1}$, this gives B_p. Since $|k_\pi|^2 \leq 3H^2$ and $|\beta_i| \geq \lambda$, Lemma 9.3 gives $|\zeta_i| \leq (8/3)H/\lambda \leq 3H/\lambda$.

For the curvature, $W = (D_{12}, -D_{02}, D_{01}) = (d_0, d_1, d_2) \times (k_0, k_1, k_2)$, and $p_\pi(d_0, d_1, d_2) = \mathcal{L} d_\pi$. Let $\mathrm{Cof}(\mathcal{L})$ have rows $L_1 \times L_2$, $L_2 \times L_0$, $L_0 \times L_1$; its determinant is $(\det \mathcal{L})^2 \neq 0$. The Binet–Cauchy identity $(L \cdot x)(L' \cdot y) - (L' \cdot x)(L \cdot y) = (L \times L') \cdot (x \times y)$, applied to pairs of rows of $\mathcal{L}$, gives $(\mathcal{L}x) \times (\mathcal{L}y) = \mathrm{Cof}(\mathcal{L})(x \times y)$, so $p_\pi^2 W = \mathrm{Cof}(\mathcal{L})\, w$. Lemma 9.7 gives $Vc = (\beta_1\, \beta_2\, \beta_3)^{\mathsf{T}} w$. Together, $c = p_\pi^2 V^{-1} (\beta_1\, \beta_2\, \beta_3)^{\mathsf{T}} \mathrm{Cof}(\mathcal{L})^{-1} W$, which defines G_p. Finally $|w|^2 \leq 3S'^2$ by Lemma 9.8, and Lemma 9.7 gives $3\lambda^2 |c_i| \leq 8S'$. □

9.4. **The lattice graph.** On a retained nonexceptional component C of degree $d \geq 4$ of a pattern p, regard $c = G_p W$ as a vector of rational functions on C, and let $\Gamma'_{m,n}$ be the image of the graph of

$$\big(s_1, s_2, s_3, c_1^m, c_2^m, c_3^m, J^n\big)$$

under the linear map that replaces the nodes s by $\zeta = B_p^{-1} s$.

Lemma 9.10 (Degree of the lattice graph)**.** *The degree δ' of $\Gamma'_{m,n}$ satisfies $h \leq \delta' \leq \big(247(m+n)+1\big)d$.*

Proof. An invertible linear change of the node coordinates is a linear automorphism of $\mathbb{A}^7$, so it preserves the affine Hilbert function, and we may keep the nodes s. The proof of Lemma 8.12, with Appendix B, uses the curvature coordinates only through the following facts.

(1) At every point P over infinity, $\operatorname{ord}_P W_j \geq 3e_P/2 > 0$ for each j (Corollary 8.8).
(2) The pole envelope $\mathcal{F}$ is supported over the affine part of C, and $\deg \mathcal{F} \geq 3d/2 + 1$ (Lemma 8.11).
(3) Some rational linear combination of the seven graph coordinates has pole divisor $\frac{5n}{2}E_\infty + m\mathcal{F}$; its construction uses only this envelope.
(4) The W_j have a common denominator Ω of degree at most 244, with numerators of degree at most 244 (Appendix B).

Each of these is unchanged when W is replaced by $c = G_p W$ for a constant invertible matrix G_p. Each c_j is a constant combination of the W_l, and each W_l is a constant combination of the c_j, so for every point P

$$\min_j \operatorname{ord}_P c_j = \min_l \operatorname{ord}_P W_l.$$

This gives (1) for c, and it shows that the pole envelope of c is $\mathcal{F}$, which gives (2) and (3). The numerators of c over Ω are constant combinations of those of W, which gives (4). The proof of Lemma 8.12 therefore applies unchanged. □

9.5. **The window bound and the optimized exponent.**

Proposition 9.11 (Lattice window bound)**.** *Fix positive integers m, n and $\eta > 0$. For a window run in $[A, 2A)$ with $T \geq 36$ satisfying* (19)*,*

$$T \ll_{m,n,\eta} a^\eta \left(X^5 + X^4 H^{1/2} + X^{5-1/(3h)} H^{1/h} + X^{5-2m/(3h)} {S'}^{m/h} + X^5 Y^{n/h} \right).$$

Proof. The proof of Proposition 8.13 applies, except for the remaining components of the nonexceptional patterns. For such a component of a pattern p, the good starts give integer points $(\zeta, c_1^m, c_2^m, c_3^m, J^n)$ of $\Gamma'_{m,n}$, and distinct node triples give distinct points because ζ determines s. By Lemma 9.9 they lie in the box of side

$$\mathcal{B}_p = \max\Big(1, \frac{3H}{\lambda(p)}, \Big(\frac{8S'}{3\lambda(p)^2}\Big)^m, Y^n\Big).$$

By Lemma 9.10 and (24), there are $O_{m,n,\eta}(a^\eta \mathcal{B}_p^{1/h})$ of them, and

$$\mathcal{B}_p^{1/h} \leq 1 + 3H^{1/h}\lambda(p)^{-1/h} + 8{S'}^{m/h}\lambda(p)^{-2m/h} + Y^{n/h}.$$

Sum over the $O_\eta(a^\eta X^5)$ patterns. Since $h \geq 2m$, both $1/h$ and $2m/h$ lie in $[0, 1]$, and Lemma 9.6, with $\tau_0 = a^{o(1)}$, gives $\sum_p \lambda(p)^{-1/h} \ll a^\eta X^{5-1/(3h)}$ and $\sum_p \lambda(p)^{-2m/h} \ll a^\eta X^{5-2m/(3h)}$. □

Corollary 9.12. *For fixed positive integers m, n and every $\varepsilon > 0$,*

$$P(a, b) \ll_{m,n,\varepsilon} a^{E'(m,n)+\varepsilon}, \qquad E'(m, n) = \max\left(\frac{5}{17}, \frac{45h}{153h - 1}, \frac{45h - 3m}{153h - 11m}, \frac{15h + 6n}{51h + 20n} \right),$$

where $h = 10n + 7m$. In particular $E'(3, 2) = 153/520$.

Proof. Argue as in the proof of Corollary 8.14, with $S' \ll a/T^3$ in place of S in (26). The first two terms and the windows in which (19) fails give the exponent $5/17$. For the third term, $T^{3h} \ll a^{O(\eta)} X^{15h-1} H^3$, and (26) gives $T^{153h-1} \ll a^{45h+O(\eta)}$. For the fourth, $T^{3h} \ll a^{O(\eta)} X^{15h-2m} {S'}^{3m}$ gives $T^{153h-11m} \ll a^{45h-3m+O(\eta)}$. For the fifth, $T^h \ll a^{O(\eta)} X^{5h} Y^n$

gives $T^{51h+20n} \ll a^{15h+6n+O(\eta)}$. For $(m,n) = (3,2)$, $h = 41$ and the last three fractions are 1845/6272, 153/520 and 627/2131. □

Proof of Theorem 1.2. Set $r = n/m$. The last two fractions in $E'(m,n)$ become

$$E'_S(r) = \frac{450r + 312}{1530r + 1060}, \qquad E_Y(r) = \frac{156r + 105}{530r + 357},$$

where E_Y is the same function as in the proof of Theorem 1.3. The first decreases and the second increases on $r > 0$. Their crossing satisfies $15r^2 = 7$, so $r_* = \sqrt{105}/15$, and their common value is $\gamma_* = 471/1559 - 6\sqrt{105}/7795$. The fraction $45h/(153h-1)$ decreases in h, and it is below γ_* once $h > \gamma_*/(153\gamma_* - 45) \approx 17.2$; also $5/17 < \gamma_*$. Choose a positive rational r close to r_* with $\max(E'_S(r), E_Y(r)) < \gamma_* + \varepsilon/2$, write $r = n_0/m_0$, and take $(m,n) = (6m_0, 6n_0)$; any scaling with $h \geq 18$ works, and the Lean proof uses 6. Then $h = 6(10n_0 + 7m_0) \geq 102$, and the fraction $45h/(153h-1)$ is at most $4590/15605 < \gamma_*$. So $E'(m,n) < \gamma_* + \varepsilon/2$, and Corollary 9.12 with loss $\varepsilon/2$ gives the theorem. □

10. Exact polar projection and the branch sieve

The curvature vector has an affine-linear projective direction. This gives a better graph degree once two possible obstructions are controlled: cancellation at its unique base point, and components of degree less than twenty-four. A local intersection calculation handles the first obstruction; a sieve for branches at infinity handles the second. We then count the additional exceptional patterns and obtain Theorem 1.1.

10.1. **An exact identity at orbit states.** For this section abbreviate the normalized parameters of (10) by

$$(\rho, u, v) = (\bar{N}_1, \bar{u}_1, \bar{u}_2), \qquad (\rho', u', v') = (\bar{N}_2, \bar{v}_1, \bar{v}_2).$$

Thus $\rho, u > 0 > v$ and $\gamma = (\rho, -\rho - u, u - v, v - 1, 1)$; primes indicate the next pattern. We retain the notation N_0, Ψ, P_4, R_4 and D_{jl} from Section 8. Put $x = (k_0, k_1, k_2)^T$ and

$$G = \begin{pmatrix} \rho(\rho+1) & -\rho(u-1) & -\rho(v-1) \\ -\rho(u-1) & u(u-1) & u(v-1) \\ -\rho(v-1) & u(v-1) & v(v-1) \end{pmatrix}, \qquad \ell = \begin{pmatrix} \rho(v-v') \\ \rho' - \rho - u(v-v') \\ u - u' - v(v-v') \end{pmatrix}.$$

In these coordinates $f = x^T G x + \ell^T x$, and $\det G = \rho(\rho+u)(u-v)(v-1) \neq 0$. Write $W = (D_{12}, -D_{02}, D_{01})^T = d \times x$.

Proposition 10.1 (Polar identity)**.** *At every orbit state with initial index i satisfying $i + 7 < P(a,b)$,*

$$\Psi W = N_0(Gx + \ell), \qquad JW = N_0^2(Gx + \ell).$$

In particular, $[W] = [Gx + \ell]$.

Proof. Set $V = Gx + \ell$. The quadratic equation gives $x \cdot V = 0$. Multiplying the five division equations by γ_j and summing, the relations $\sum \gamma_j = \sum \gamma_j s_j = \sum \gamma_j b_j = 0$ leave

$$\sum_{j=0}^{4} \gamma_j s_j t_j + \sum_{j=0}^{4} \gamma_j d_j = 0, \qquad t_j = b_0 - b_j.$$

The cumulative drop and gap identities identify this expression with $d \cdot V$. For $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$, expansion also gives

$$V \cdot (r_0, r_1, r_2) - \Psi = -(s_4 + 1 + v - v')f.$$

Thus $V \cdot r = \Psi$, whereas $W \cdot r = N_0$. Since $N_0 > 0$, the vectors d, x are independent. Their common orthogonal line contains both V and W, so $\Psi W = N_0 V$. Now use $J = N_0 \Psi$ and $\Psi > 0$. □

The three homogeneous forms $Gx + \ell w$ have a unique common zero, the finite point $x_* = -G^{-1}\ell$. It lies on $f = \Psi = 0$. It also lies on the fixed-dividend curve for every a. Indeed, its nodes are

$$s_0^* = 0, \qquad s_j^* = \frac{\gamma'_{j-1} - \gamma_{j-1}}{\gamma_j} \ (1 \le j \le 3), \qquad s_4^* = v' - v.$$

Dividing $Z(\xi) = \xi R_4(\xi)$ by $P_4(\xi)$ gives quotient $\xi + h$, where $h = v' - v - s_4$, and a remainder with Newton coefficients

$$-h\gamma_0, \qquad \gamma'_{j-1} - \gamma_{j-1} - \gamma_j(s_j + h) \quad (1 \le j \le 3).$$

They all vanish at x_*, so $Z_* = \xi P_{4,*}$ and $N_0 \operatorname{Res}(P_{4,*}, Z_*) - a\Psi(x_*)^4 = 0$.

We add the three denominator-cleared polar identities to the retained cuts of Section 8.2. Each has degree at most 250. A cut that is not identical on a degree-d component removes all but $250d$ of its points, by Bézout's theorem. Hence these additional cuts cost $O(1)$ points per pattern. Throughout this section, a *retained* component satisfies the earlier cuts and these three new ones; on it,

$$JW = N_0^2(Gx + \ell)$$

holds in the function field.

10.2. **The base point and the graph degree.**

Lemma 10.2 (The common-base divisor)**.** *There is a fixed rational polynomial Θ in the ten minor coordinates, nonzero on the decomposable cone, with the following property. For every pattern with $\Theta \neq 0$ and every integral component C of the pattern curve, the common-base divisor of the homogeneous forms $Gx + \ell w$ on the normalization of C has degree at most four. The polynomial Θ is independent of a.*

Proof. Work first where $\ell_0 \neq 0$. Since $\nabla f(x_*) = -\ell$, the quadric is smooth at x_*. Take $s = x_1 - x_{*,1}$ and $t = x_2 - x_{*,2}$ as local coordinates and solve $f = 0$ for x_0 by formal power series. Let $R_1(\xi; s, t)$ be the linear part of the Newton remainder just computed, and let $\Psi_1(s, t)$ be the linear part of Ψ on this tangent plane. The curve equation has order at least four and quartic initial form

$$N_0 N_4(s,t) - a\Psi_1(s,t)^4, \qquad N_4(s,t) = \operatorname{Res}_\xi(P_{4,*}, R_1). \tag{27}$$

To see that only first derivatives enter, compute the resultant as a 4×4 Bézout determinant. Each row vanishes at x_*, so its term of degree four uses exactly the linear part of every row. Higher terms in the formal solution of $f = 0$ cannot change (27).

Consider the rational function of the pattern parameters

$$M = N_4(1,0)\Psi_1(0,1)^4 - N_4(0,1)\Psi_1(1,0)^4.$$

If $M \neq 0$ and $N_0 \neq 0$, the two axis coefficients in (27) cannot both vanish, for any a. Appendix D gives the tangent frame, quartic and first derivatives at

$$(\rho, u, v, \rho', u', v') = (2, 5, -3, 3, 6, -5), \qquad x_* = (-1/7, -3/28, -1/2),$$

where

$$M = \frac{26291112152499}{60236288} \neq 0.$$

These parameters arise from the maximal minors of the integer matrix

$$\begin{pmatrix} 1 & 0 & 0 & 2 & 10 \\ 0 & 1 & 0 & -5 & -22 \\ 0 & 0 & 1 & 3 & 9 \end{pmatrix}.$$

Clear denominators in M and the chart conditions, and include the necessary nonzero denominator factors. Their product defines Θ. The displayed matrix witnesses its nonvanishing on the decomposable cone.

We now justify the passage from a quartic initial term to a divisor bound. Let S be the local ring of the smooth quadric at x_* and let F be the curve equation. Choose $h = s$ or $h = t$ so that the complementary axis in (27) has nonzero coefficient. The completion of $S/(h)$ is a power-series ring in one variable, in which F has order exactly four. Consequently $\dim_{\mathbb{Q}} S/(F, h) = 4$. If C passes through x_*, its local ring $R = S/I_C$ is a quotient with $F \in I_C$, so

$$\dim_{\mathbb{Q}} R/hR \le 4.$$

In particular, h is nonzero on this one-dimensional integral local ring.

Let $\overline{R}$ be its finite normalization and $Q = \overline{R}/R$. The vector space Q is finite-dimensional over $\mathbb{Q}$. Apply multiplication by h to $0 \to R \to \overline{R} \to Q \to 0$. Multiplication is injective on the first two rings, and its kernel and cokernel on Q have equal dimensions. The resulting exact sequence therefore gives

$$\dim_{\mathbb{Q}} \overline{R}/h\overline{R} = \dim_{\mathbb{Q}} R/hR = \sum_{P|x_*} [\kappa(P) : \mathbb{Q}] \operatorname{ord}_P h.$$

The final equality follows by factoring the ideal (h) in the semilocal Dedekind ring $\overline{R}$. Since G is invertible, the coordinates of $Gx + \ell = G(x - x_*)$ generate the centered maximal ideal. Their minimum order at each P is at most $\operatorname{ord}_P h$. The weighted sum of these minimum orders is therefore at most four. A component missing x_* has no common-base contribution. □

Suppose now that a retained nonexceptional component has degree 24 and $\Theta \neq 0$. At each point at infinity, $\operatorname{ord}_P J = -5e_P/2$, by Corollary 8.4, and $\sum e_P[\kappa(P) : \mathbb{Q}] = 24$. Thus J has pole divisor, and hence zero divisor, of degree 60; all its zeros are affine. In the identity $W = N_0^2(Gx + \ell)/J$, cancellation at the affine zeros of J can cost at most the common-base degree. Lemma 10.2 therefore gives an affine curvature pole envelope of degree at least $60 - 4 = 56$. These poles are disjoint from the poles of J at infinity, so the pole formula of Lemma 8.12 gives

$$\deg(\text{powered joint graph}) \ge h := 60n + 56m. \tag{28}$$

An invertible constant change of curvature coordinates preserves the minimum valuation, so this also holds for the lattice coordinates of Section 9. It remains to show that components of degree less than 24 occur for relatively few patterns.

10.3. **An explicit label for a branch at infinity.** Let P be a normalized branch at infinity of a nonexceptional component. The five leading nodes are distinct by Lemma 8.3. Scale them to $(0, 1, t, y, z)$, and put

$$z = -\gamma_1 - \gamma_2 t - \gamma_3 y.$$

The second and third moment equations become

$$Q_2(t, y) = \gamma_1 + \gamma_2 t^2 + \gamma_3 y^2 + z^2 = 0, \qquad Q_3(t, y) = \gamma_1 + \gamma_2 t^3 + \gamma_3 y^3 + z^3 = 0.$$

Define the universal sextic $E_\gamma(t)$ by the polynomial identity

$$\operatorname{Res}_y(Q_2, Q_3) = \gamma_3^2(\gamma_3+1)E_\gamma(t). \tag{29}$$

The quotient on the right is a polynomial over $\mathbb{Z}[\gamma_1,\gamma_2,\gamma_3]$, of degree six in t. At an orbit pattern $\gamma_3 = v-1 < -1$, so $\gamma_3^2(\gamma_3+1) \neq 0$. Cancelling this factor in the resultant and linear subresultant identities gives $E_\gamma(t) = 0$ and $d(t)y + n(t) = 0$, where

$$\begin{aligned} d(t) &= (\gamma_3-1)\big(\gamma_2(\gamma_2+\gamma_3+1)t^2 + 2\gamma_1\gamma_2 t + \gamma_1(\gamma_1+\gamma_3+1)\big),\\ n(t) &= \gamma_2(\gamma_2+1)(\gamma_2+\gamma_3+1)t^3 + \gamma_1\gamma_2(3\gamma_2+\gamma_3+2)t^2\\ &\quad + \gamma_1\gamma_2(3\gamma_1+\gamma_3+2)t + \gamma_1(\gamma_1+1)(\gamma_1+\gamma_3+1). \end{aligned}$$

These are polynomial identities, so the cleared relation remains valid when a specialized denominator vanishes.

Write

$$\Delta = ty(t-1)(y-1)(y-t), \qquad V_5 = z(z-1)(z-t)(z-y).$$

By (16), the degree-ten leading resultant at these nodes is $R_0 = \gamma_0\gamma_1\gamma_2\gamma_3\Delta V_5$. The four moment equations give the barycentric identity

$$R_0\Delta = V_5^4. \tag{30}$$

For example, the identities $\gamma_j\prod_{l\neq j}(z_j - z_l) = V_5$ for $0 \le j < 4$ follow by evaluating the first four moment relations; their product is (30). Let $w = 1/s_1$ on this chart. The residue $\eta_P = \operatorname{res}_P(w^5\Psi^2)$ is nonzero, because $\operatorname{ord}_P\Psi = -5e_P/2$. Taking leading terms in the curve equation gives $N_0R_0 = a\eta_P^2$. Thus the actual residue field contains

$$\beta = V_5^2/\eta_P, \qquad N_0\beta^2 = a\Delta. \tag{31}$$

This quadratic relation supplies information beyond the six possible locations t.

For a slope c define

$$D_\gamma(t) = t(-n(t))(t-1)(-n(t)-d(t))(-n(t)-td(t))$$

and the label polynomial

$$L_{a,N_0,c,\gamma}(Z) = \operatorname{Res}_t\big(E_\gamma(t), N_0d(t)^3(Z-t)^2 - c^2aD_\gamma(t)\big). \tag{32}$$

Here the resultant means the Sylvester determinant with degree bounds 6 and 11, padding missing coefficients with zero. This convention makes (32) a polynomial family under every specialization. The relation $d(t)y + n(t) = 0$ gives $D_\gamma(t) = d(t)^3\Delta$, so (31) implies

$$L_{a,N_0,c,\gamma}(t + c\beta) = 0$$

in every actual residue field, including when $d(t) = 0$.

Lemma 10.3 (The degree-twelve label)**.** *Fix $a \in \mathbb{Q}^\times$, and regard $N_0,\gamma_1,\gamma_2,\gamma_3$ as independent parameters. For some integer c with $0 \le c \le 144$, $L_{a,N_0,c,\gamma}$ is irreducible of degree twelve over $\mathbb{Q}(N_0,\gamma_1,\gamma_2,\gamma_3)$. For a specialized nonexceptional pattern, if one such label polynomial is irreducible of degree twelve over $\mathbb{Q}$, every integral component of its pattern curve has degree twenty-four.*

Proof. At $(\gamma_1,\gamma_2,\gamma_3) = (-7,8,-4)$, the sextic in (29) equals $24E_0(t)$, where

$$E_0(t) = 2400t^6 - 20160t^5 + 69776t^4 - 126448t^3 + 125636t^2 - 64680t + 13475.$$

Appendix D proves E_0 irreducible modulo 71. This specialization preserves degree six. Gauss's lemma therefore shows that E_γ is irreducible over $K = \mathbb{Q}(\gamma_1,\gamma_2,\gamma_3)$: a factorization into two factors of positive t-degree over the parameter ring would specialize to two factors of positive

degree. Choose a root t of E_γ, so $[K(t):K]=6$. In this field $d(t)\neq 0$ since its degree is less than six, and $y=-n(t)/d(t)$. The same degree comparison shows $\Delta\neq 0$.

Adjoin the independent parameter N_0. The element $a\Delta/N_0$ has odd valuation at $N_0=0$, so it is not a square in $K(t)(N_0)$. Thus (t,β) generates an extension of degree twelve over $K(N_0)$. Among its twelve embeddings, equality of two values $t+c\beta$ excludes at most one slope per pair. Consequently one of $0,\dots,144$ gives a primitive element. Its minimal polynomial divides (32), whose degree is at most twelve; hence that label polynomial is irreducible of degree twelve. This argument holds for each fixed nonzero a; the chosen slope may depend on a.

For an actual pattern, the residue field at any infinity branch contains the root $t+c\beta$. Irreducibility of the specialized degree-twelve label forces $[\kappa(P):\mathbb{Q}]\geq 12$. Corollary 8.4 gives $e_P\geq 2$, so

$$\deg C=\sum_{P\text{ at infinity}} e_P[\kappa(P):\mathbb{Q}]\geq 24.$$

Lemma 8.1 gives the reverse inequality. □

Call a pattern *branch-good* if some rational slope c makes its label irreducible of degree twelve; otherwise call it *branch-bad*. Only the bounded set of integer slopes from Lemma 10.3 will be needed to estimate the latter set.

10.4. **The arithmetic sieve.** We use the following monic variant of effective Hilbert irreducibility. Its proof follows the approach of [4, Theorem 1.10]; the logarithmic powers below are those of the monic variant proved here.

Theorem 10.4 (Monic specialization estimate)**.** *For fixed integers $r,d\geq 1$ there is a constant $C_{r,d}$ with the following property. Let $F\in\mathbb{Z}[T_1,\dots,T_r][Y]$ be monic in Y and irreducible over $\mathbb{Q}(T_1,\dots,T_r)$. Suppose $\deg_Y F\leq d$, each coefficient of F has total parameter degree at most d, and every integer coefficient has absolute value at most $H\geq 1$. For every integer $B\geq 1$,*

$$\#\{t\in[-B,B]^r\cap\mathbb{Z}^r : F(t,Y)\text{ is reducible over }\mathbb{Q}\}\leq C_{r,d}(1+\log(H+1))^4B^{r-1/2}(1+\log B)^{4r+1}.$$

Proof. First consider an irreducible $A(T,Z)\in\mathbb{Z}[T,Z]$ of bounded degree, height H_0, and Z-degree $e\geq 2$. Apart from the bounded number of zeros of its leading coefficient, an integer root at $|T|\leq B$ has $|Z|\ll_d H_0B^d$, by the root bound. Apply the shear $Z\mapsto Z+T^k$, with $k\gg_d 1+\log(H_0+1)$. The transformed irreducible curve has degree between ke and $kd+d$, and the transformed points have height $O_d(H_0B^k)$, taking $k>d$. The coefficient-uniform curve bound (24) consequently gives

$$\#\{t\in[-B,B]\cap\mathbb{Z} : A(t,Z)\text{ has an integer root}\}\ll_d(1+\log(H_0+1))^4B^{1/e}(1+\log B).$$

Indeed, $H_0^{1/(ke)}$ is bounded, and the degree and logarithmic factors cost $O_d(k^4(1+\log B))$. The case $B=1$ is immediate.

For a monic $F(T,Y)$, a proper factor of a specialization is monic with integer coefficients, by Gauss's lemma. Its coefficients are elementary symmetric functions of a proper subset of the roots. For each generic proper subset, at least one such coefficient is outside $\mathbb{Q}(T)$, or the generic polynomial would factor. Its irreducible coefficient equation has root degree at least two. Universal symmetric resolvents bound the number and degrees of these equations in terms of d; integer factor-height bounds bound their heights by $C_dH^{C_d}$. Specialize in the integral closure of $\mathbb{Z}[T]$ in a splitting field. Every specialized proper factor then supplies an integer root of one of these equations, even when specialized roots coincide. The preceding root estimate and a finite union give the one-parameter bound.

For one parameter with remaining polynomial variables, use Kronecker substitution: encode the remaining variables and Y by successive powers of one variable, with base larger than their

degree bounds and with Y in the highest position. The encoded polynomial is still monic, with degree bounded in terms of r, d. Factor it over $\mathbb{Q}(T_1)$. Outside the one-parameter exceptional set, its irreducible factors remain irreducible after specialization. A factorization of the original specialization must therefore come from one of finitely many products of these factors. Decode each candidate pair. If its product were the original family identically, it would contradict generic irreducibility; otherwise one nonzero coefficient polynomial in T_1 must vanish. There are only $O_{r,d}(1)$ such extra parameter values. This proves the same square-root bound for failure of irreducibility as a family in the remaining variables.

Finally induct on r. Exceptional values of T_1 contribute their count times $(2B+1)^{r-1}$. At each remaining value, apply the induction hypothesis to the specialized family, whose coefficient height is at most $(d+1)HB^d$. Since

$$1+\log((d+1)HB^d+1) \ll_d (1+\log(H+1))(1+\log B),$$

each induction step costs at most four further logarithmic powers. The stated exponent $4r+1$ bounds all these losses. □

To apply the theorem with a fixed, use the four integer minors of the first four columns,

$$\mathbf{v} = (v_0, v_1, v_2, v_3) = (p_{012}, p_{013}, p_{023}, p_{123}).$$

For $v_0 = N_0 \neq 0$ their relation to the independent parameters above is

$$\gamma_1 = -\frac{v_3+v_2}{v_0}, \qquad \gamma_2 = \frac{v_2+v_1}{v_0}, \qquad \gamma_3 = -\frac{v_1+v_0}{v_0}. \tag{33}$$

This change is birational. Substitute it in (32) and clear denominators by a fixed power of v_0. To make the resulting degree-twelve polynomial monic without changing its generic irreducibility, use the following polynomial operation. For $F(Y) = \sum_{i=0}^{q} b_i Y^i$ put

$$\mathcal{M}_q(F)(Y) = Y^q + \sum_{i<q} b_i b_q^{q-1-i} Y^i.$$

When $b_q \neq 0$, this is $b_q^{q-1} F(Y/b_q)$. When $b_q = 0$ and $q \geq 2$, it is $Y^{q-1}(Y + b_{q-1})$, hence reducible. Thus irreducibility of a specialized monic normalization implies both preservation of degree and irreducibility of the original label.

For each fixed positive integer a, choose the slope supplied by Lemma 10.3 and set $q = 12$. We obtain a generically irreducible monic integer family in $\mathbf{v}$, with uniformly bounded degree and coefficient height at most $C(a+1)^D$, for absolute constants C, D. The same is true after dilation $\mathbf{v} = g\mathbf{z}$, with height at most $C(a+1)^D g^D$. The slope may depend on a, but the bounds are uniform because there are only 145 choices. Theorem 10.4, together with the $O(B^3)$ zero-scale slice $v_0 = 0$, gives

$$\#\{\mathbf{z} \in [-B, B]^4 \cap \mathbb{Z}^4 : g\mathbf{z} \text{ is branch-bad}\} \ll_\eta a^\eta B^{7/2} \tag{34}$$

whenever $1 \leq g, B \ll a^7$. Vertices with $v_0 = 0$ are included among the bad vertices. This range contains all the dilations and boxes used for orbit patterns below.

Lemma 10.5 (Branch-bad full patterns)**.** *Among the selected orbit patterns of coordinate height at most X, the number that are branch-bad is $O_\eta(a^\eta X^{9/2})$.*

Proof. Partition the first-four-column vertices into dyadic bands with $\max_j |v_j| \asymp M$, and choose a maximal coordinate as signed pivot q. The other three vertex coordinates form u; put $g = \gcd(q, u_1, u_2, u_3)$. Completing the fifth column amounts to choosing $v \in [-X, X]^3$ with $q \mid u \times v$. The congruence count is

$$O(X^3 g^2 / M^2).$$

In particular, these completion coordinates remain in the full X-box, even when M is small. Apply (34) after dilation by g. There are $O_\eta(a^\eta(M/g)^{7/2})$ bad vertices in this common-factor class; imposing the exact gcd only reduces this count. Their total contribution is at most

$$O_\eta\left(a^\eta X^3 M^{3/2}\sum_{g\ge 1} g^{-3/2}\right).$$

The series converges, and summing the bands $M \le X$ proves the claim. The finitely many pivot choices only change the constant. □

10.5. **Counting the geometric exceptional locus.** The new hypersurface $\Theta = 0$ requires a saving over the ordinary $O(X^5)$ pattern count. The following estimate includes all common factors.

Proposition 10.6 (Hypersurface pattern count)**.** *Let Θ be a rational polynomial in the ten minor coordinates, with $\Theta(\operatorname{plk} M) \neq 0$ for some rational 3×5 matrix M. For every $\eta > 0$, the number of integral Plücker vectors of coordinate height at most $X \ge 1$ satisfying $\Theta = 0$ is*

$$O_{\Theta,\eta}(X^{14/3+\eta}).$$

The count includes nonprimitive vectors and the zero vector.

Proof. Choose a minor of maximal absolute value $q > 0$ and record its sign. The six one-column replacements form $u, v \in [-q, q]^3$, and the other coordinates are signed entries of $(u \times v)/q$. There are twenty signed pivot charts. In each chart, clearing denominators in Θ gives a nonzero polynomial $B(q, u, v)$. Indeed, the product of the pivot minor and $\Theta(\operatorname{plk} M)$ is a nonzero polynomial in the matrix entries, so it is nonzero at some rational matrix; every pivot chart meets the nonvanishing locus.

Put $g = \gcd(q, u_1, u_2, u_3)$, $n = q/g$ and $u_0 = u/g$. Since $\gcd(n, u_0) = 1$, the congruence $u_0 \times v \equiv 0 \pmod n$ is equivalent to

$$v \in \Lambda := \mathbb{Z}u_0 + n\mathbb{Z}^3.$$

Let $\mu(u_0)$ be the least nonzero sup norm in the normal lattice $\Lambda_n^\perp = \{z \in \mathbb{Z}^3 : n \mid z \cdot u_0\}$. If $x, y \in \Lambda$ are independent, then $x \times y = nz$ for a nonzero $z \in \Lambda_n^\perp$. Hence $|x||y| \gg n\mu(u_0)$. For a reduced basis of Λ with lengths ℓ_1, ℓ_2, ℓ_3, this gives $\ell_i\ell_j \gg n\mu(u_0)$ for each pair. The basis lengths are $O(n)$: apply the coordinate bound of Lemma 9.3 to n times each standard basis vector, which lies in Λ. For each basis coordinate, at least one of these three integer coordinates is nonzero. By the coordinate bounds of Lemma 9.3, vectors with $|v|_\infty \le q$ have basis coordinates in a box of side lengths $O(q/\ell_i)$; these lengths are bounded below because $q \ge n$. For a nonzero polynomial fiber $B(q, u, \cdot)$, the polynomial grid-zero bound therefore gives

$$\#\{v \in \Lambda \cap [-q, q]^3 : B(q, u, v) = 0\} \ll_B \sum_{i<j} \frac{q^2}{\ell_i\ell_j} \ll_B \frac{q^2}{n\mu(u_0)}. \tag{35}$$

Here is the reciprocal estimate that produces the exponent $14/3$:

$$\sum_{u_0\in[-n,n]^3\cap\mathbb{Z}^3} \frac{1}{\mu(u_0)} \ll \tau(n)n^{8/3}. \tag{36}$$

It holds for all u_0, without a primitivity restriction. Pigeonholing $z \mapsto u_0 \cdot z \pmod n$ gives $\mu(u_0) \ll n^{1/3}$. For $R \geq 1$, Lemma 9.4 and a divisor sum give

$$\begin{aligned}\#\{u_0 : \mu(u_0) \leq R\} &\leq \sum_{0<|z|_\infty \leq R} \#\{u_0 \in [-n,n]^3 : n \mid u_0 \cdot z\} \\ &\ll n^2 \sum_{0<|z|_\infty \leq R} \gcd(n, z_1, z_2, z_3) \ll \tau(n) n^2 R^3.\end{aligned}$$

For the last bound, expand the gcd as a divisor sum and note that a nonzero z of size at most R has no common divisor greater than R. Partial summation up to $K \asymp n^{1/3}$ now bounds the reciprocal sum by $O(n^3/K + \tau(n) n^2 K^2)$, proving (36).

For fixed q, g, equations (35) and (36) bound the nonzero fibers by $O_B(\tau(n) q^2 n^{5/3})$. Sum over $g \mid q$ and $q \leq X$ to obtain

$$O_B\big(\tau_{\max}(X)^2 X^{14/3}\big), \qquad \tau_{\max}(X) = \max_{1 \leq j \leq X} \tau(j).$$

Finally, an identically zero fiber forces one fixed nonzero coefficient polynomial $B_0(q,u)$ to vanish. For fixed g, the polynomial $B_0(gq', gu')$ is still nonzero. The four-variable grid-zero bound supplies $O_B((X/g)^3)$ possible scaled vertices. Each has $O(qg^2) \leq O(Xg^2)$ completions in its q-box, so these fibers total

$$O_B\left(X^4 \sum_{g \leq X} \frac{1}{g}\right) = O_B(X^4 \log(2X)).$$

The divisor estimate absorbs $\tau_{\max}(X)^2$ into X^η. Summing the charts and adding the zero vector completes the proof. □

10.6. **The window estimate and the polar exponent.** Retain the window notation $[A, 2A)$ of the earlier sections. For $T \geq 36$ consecutive positive terms, put

$$H = \left\lfloor \frac{28a}{AT} \right\rfloor + 1, \quad X = \left\lfloor \frac{3609600 A^4 H^6}{a^3} \right\rfloor, \quad S' = \left\lfloor \frac{9472 A^3 H^3}{a^2} \right\rfloor, \quad Y = \left\lfloor \frac{1259991040 A^5 H^{10}}{a^4} \right\rfloor.$$

Only windows satisfying (19) need the new estimate. As before, call $R : \mathbb{N} \to \mathbb{N}$ slowly growing if $R(a) \ll_\eta a^\eta$ for every $\eta > 0$.

Proposition 10.7 (Polar window estimate)**.** *For fixed positive integers m, n, set $h = 60n + 56m$. There is a slowly growing R, depending only on m, n, such that every window above satisfying* (19) *obeys*

$$\begin{aligned}T \leq R(a)\big(&X^5 + X^4 H^{1/2} + X^{14/3} \max(H, S', Y)^{1/17} \\ &+ X^{5-1/(3h)} H^{1/h} + X^{5-2m/(3h)} (S')^{m/h} + X^5 Y^{n/h}\big).\end{aligned} \tag{37}$$

Proof. The earlier exceptional patterns, conics and bounded cuts contribute $X^5 + X^4 H^{1/2}$, as in Proposition 9.11. Among the remaining patterns, Lemma 10.5 and Proposition 10.6 bound those that are branch-bad or satisfy $\Theta = 0$ by $O_\eta(a^\eta X^{14/3})$. Here $X \ll a^7$, so the extra X^η loss is absorbed into a^η. Use the earlier degree-17 unpowered graph for these patterns, giving the third term of (37).

For each remaining retained component, Lemma 10.3 gives degree 24, and (28) gives graph degree at least h. Its upper degree is bounded in terms of m, n by Lemma 8.12. The coefficient-uniform curve bound and Lemma 9.9 therefore give, for a pattern with shortest lattice length

λ, a contribution bounded by a slowly growing factor times

$$(H/\lambda)^{1/h} + (S'/\lambda^2)^{m/h} + Y^{n/h}.$$

Summing with Lemma 9.6, at exponents $1/h$ and $2m/h$, produces the last three terms. All omitted constants and logarithmic factors depend only on the fixed powers and fit into R. □

Theorem 10.8 (Polar bound)**.** *For every fixed pair of positive integers m, n and every $\varepsilon > 0$,*

$$P(a,b) \ll_{m,n,\varepsilon} a^{E_{\mathrm{pol}}(m,n)+\varepsilon} \quad (0 < b < a),$$

where, with $h = 60n + 56m$,

$$E_{\mathrm{pol}}(m,n) = \max\left\{\frac{5}{17}, \frac{45h}{153h-1}, \frac{45h-3m}{153h-11m}, \frac{15h+6n}{51h+20n}\right\}.$$

For $(m,n) = (3,2)$ this exponent is $4317/14677$. Optimizing over the fixed powers gives

$$P(a,b) \ll_{\varepsilon} a^{\beta_*+\varepsilon}, \qquad \beta_* = \frac{41301 - 3\sqrt{82849}}{137480} = 0.2941336553\ldots.$$

Proof. Apply the window argument of Section 9 to (37). The three terms from the exceptional contribution $X^{14/3}\max(H, S', Y)^{1/17}$ give exponents $717/2440$, $717/2440$ and $732/2491$, all below $5/17$. The other terms give the displayed maximum. The six-step and drop bounds handle windows outside (19); dyadic summation and slow growth absorb the remaining losses.

For $(m,n) = (3,2)$, $h = 288$ and the last three fractions are $12960/44063$, $4317/14677$ and $1083/3682$. For optimization put $r = n/m$. The curvature and rank-five fractions become

$$E_S(r) = \frac{2700r + 2517}{9180r + 8557}, \qquad E_Y(r) = \frac{906r + 840}{3080r + 2856}.$$

The first decreases and the second increases. They cross at $r_* = (\sqrt{82849} - 47)/360$, with common value β_*. Approximate r_* by positive rationals and scale the corresponding numerator and denominator until $45h/(153h-1) < \beta_*$. The arbitrarily small exponent loss absorbs the approximation, proving Theorem 1.1. □

11. Rises in the quotient increments

Throughout this section $k_i = q_{i+1} - q_i \geq 1$, $d_i = b_i - b_{i+1} \geq 1$, and a *rise* is an index $0 \leq i \leq P(a,b) - 3$ with $k_i < k_{i+1}$.

Lemma 11.1 (Six quotients in arithmetic progression)**.** *Let $q, h \geq 1$ and $x_0 > x_1 > \cdots > x_6 \geq 0$ satisfy $x_j(q + jh) + x_{j+1} = a$ for $0 \leq j \leq 5$. Then $q^4 \leq 324\, h^3 a$.*

Proof. Let $f_j = x_j - x_{j+1}$, and let $E = (f_1 - f_2)h$ and $F = (f_2 - f_3)h$ be the curvatures of the blocks starting at x_1 and x_2 (all their increments equal h). With $U = f_0 - 2f_1 + f_2$, the division equations $\mathcal{E}_j\colon x_j(q + jh) + x_{j+1} - a = 0$ give the exact identity

$$qhU = (3h-1)E + F, \qquad \text{namely} \qquad qhU - (3h-1)E - F = h(\mathcal{E}_0 - 3\mathcal{E}_1 + 3\mathcal{E}_2 - \mathcal{E}_3).$$

This is the case $u = v = w = z = h$ of Lemma 5.1, where $L = 2h^3U$. By Lemma 3.2, $E, F \geq 1$, so the integer U is at least 1.

Now bound E and F. Subtracting consecutive division equations gives $q_j f_j \leq h x_{j+1}$, where $q_j = q + jh$. Since $x_{j+1}q_{j+1} \leq a$, this yields $q^2 f_j \leq ha$ for $1 \leq j \leq 4$. By Lemma 3.4, $Ex_3 \leq (h+1)(f_1+f_2+f_3)^2 \leq 9h^2(h+1)a^2/q^4$. Also $a < x_3(q+3h+1)$, so $Eq^4 \leq 9h^2(h+1)(q+3h+1)a$, and likewise $Fq^4 \leq 9h^2(h+1)(q+4h+1)a$. Hence

$$q^5 h \leq q^4\big((3h-1)E + F\big) \leq 27h^3(h+1)(q+4h+1)a.$$

If $q \geq h$ this gives $q^4 \leq 324h^3a$. If $q < h$ then $q^4 < h^4 \leq h^3 a$, since $h \leq q_1 \leq a$. □

Lemma 11.2 (Drops decrease between rises)**.** *Whenever $b_{i+2} > 0$, $d_{i+1}k_i < d_ik_{i+1}$. In particular, if $k_{i+1} \le k_i$ then $d_{i+1} < d_i$.*

Proof. This is $D = d_ik_{i+1} - d_{i+1}k_i \ge 1$ (Lemma 3.2). If $k_{i+1} \le k_i$ and $d_{i+1} \ge d_i$, then $d_ik_{i+1} \le d_{i+1}k_i$, which is a contradiction. □

Lemma 11.3 (Bounded multiplicity)**.** *Let $c_j \ge 1$ be integers, $j \in J$, such that no value occurs more than M times. Then $|J|^2 \le 2M \sum_{j\in J} c_j$.*

Proof. Induct on $|J|$. Let c_{j_0} be the largest value. The values lie in $[1, c_{j_0}]$, so $|J| \le Mc_{j_0}$. Removing j_0 and using $|J|^2 = (|J|-1)^2 + 2|J| - 1$ completes the induction. □

Proposition 11.4 (Rises in a window)**.** *Let $x_0, \dots, x_{T-1}$ be a window run in $[A, 2A)$ with $T \ge 2$, and let ρ be the number of rises among its $T-1$ increments. Then:*

1. *$(T-1)^2 \le 2(\rho+1)(A-1)$;*
2. *$T - 1 \le \lfloor a/A \rfloor$;*
3. *if $6A \le a$, then $T - 1 = \ell + s$ with $\ell^2 \le 8(\rho+1)\lfloor a/A \rfloor$ and $s^3 \le 26244\,A$.*

Proof. Split the increments into *runs*: maximal stretches containing no rise. There are at most $\rho + 1$ runs.

(1) By Lemma 11.2, the drops strictly decrease along a run, so each value of d_j occurs at most once per run, hence at most $\rho + 1$ times. Lemma 11.3 and $\sum_j d_j = x_0 - x_{T-1} \le A - 1$ give the claim.

(2) The increments are ≥ 1, and they sum to $q_{T-1} - q_0 \le \lfloor a/A \rfloor$.

(3) Call an increment h *low* if $26244h^3A^4 < a^3$. Five consecutive equal increments h inside the run give six quotients in arithmetic progression. Lemma 11.1 then gives $q^4 \le 324h^3a$. On the other hand, $a < 3Aq$, which follows from $a = qx_0 + x_1$, $x_1 < x_0 < 2A$ and $6A \le a$. Together these give $a^3 \le 26244h^3A^4$, so h is not low. Within a run the increments are nonincreasing, so equal values are consecutive. Hence each low value occurs at most 4 times per run, and at most $4(\rho+1)$ times in all. Lemma 11.3 bounds the number ℓ of low increments by $\ell^2 \le 8(\rho+1)\sum_j k_j \le 8(\rho+1)\lfloor a/A \rfloor$. Let s be the number of high increments. If $s = 0$, its bound is immediate. Otherwise let h_* be their smallest value. Then $sh_* \le \lfloor a/A \rfloor$ and $a^3 \le 26244h_*^3A^4$, so $s^3a^3 \le 26244A^4(a/A)^3 \le 26244Aa^3$. □

Lemma 11.5 (Geometric sums)**.** *Let $\alpha, \beta, s, t > 0$ and $K \ge 0$. Then*

$$\sum_{k<K} \min\bigl(\alpha 2^{-sk},\ \beta 2^{tk}\bigr) \le \Bigl(\frac{1}{1-2^{-s}} + \frac{2^t}{2^t-1}\Bigr)\alpha^{t/(s+t)}\beta^{s/(s+t)}.$$

Proof. Let $M = \alpha^{t/(s+t)}\beta^{s/(s+t)}$. Then $(\alpha 2^{-sk})^t(\beta 2^{tk})^s = M^{s+t}$, so for each k either $\beta 2^{tk} \le M$ or $\alpha 2^{-sk} < M$. The terms $\beta 2^{tk} \le M$ grow with ratio 2^t, so they sum to at most $M2^t/(2^t-1)$. The terms $\alpha 2^{-sk} < M$ decay with ratio 2^{-s}, so they sum to at most $M/(1-2^{-s})$. □

Proof of Theorem 1.5. The rises inside a window are rises of the orbit, so $\rho + 1 \le R$. Combining the three parts of Proposition 11.4 gives for the window $[2^k, 2^{k+1})$ (when $6A > a$, part (2) gives $T - 1 \le a/A < 6 \le 30A^{1/3}$)

$$T_k \le 1 + \sqrt{R}\,\min\bigl(\sqrt{8a}\,2^{-k/2},\ \sqrt{2}\,2^{k/2}\bigr) + \min\bigl(a\,2^{-k},\ 30\cdot 2^{k/3}\bigr).$$

Lemma 11.5 gives $\sum_k \min(\sqrt{8a}\,2^{-k/2}, \sqrt{2}\,2^{k/2}) \ll (\sqrt{8a}\sqrt{2})^{1/2} = 2a^{1/4}$. It also gives $\sum_k \min(a2^{-k}, 30\cdot 2^{k/3}) \ll a^{1/4}$. Since $\log_2 a + 1 \ll a^{1/4}$, summing over k gives the theorem. □

Remark 11.6. By Theorem 1.5, an orbit with $P(a,b) \asymp a^{3/10}$ must have $R \gg a^{1/10}$. Proposition 11.4(3) says more: inside the critical window $A \asymp a^{3/5}$, a run of $T \asymp a^{3/10}$ remainders must contain $\gg a^{1/5}$ rises.

12. The Lean formalization

The source repository is `https://github.com/jasondavies/pierce-length-bound` (private during manuscript preparation).

The proof baseline is commit `27e6ac9ed3e49378064126fbe3c80b9b1f91ffd3`. The development uses Lean 4 `v4.35.0` and Mathlib `v4.35.0`. The unconditional results of this paper are proved without additional mathematical hypotheses, including the curve-counting input and the effective Hilbert specialization theorem required by Theorem 1.1.

The strongest bound is exported as

```
theorem P_le_polar_optimal :
    ∀ ε > (0 : ℝ), ∃ C > 0, ∀ a b : ℕ,
    0 < b → b < a → (P a b : ℝ) ≤
    C * (a : ℝ) ^ (((41301 : ℝ) - 3 * Real.sqrt 82849) / 137480 + ε)
```

The quantifier order makes uniformity explicit: C may depend on ε, but not on a or b. The function `P a b` is defined by well-founded recursion,

$$P(a,0) = 0, \qquad P(a,b) = 1 + P(a, a \bmod b) \quad (b > 0),$$

and counts the positive terms of the zero-indexed orbit, as in the paper. The rational bound 4317/14677 is exported separately as `P_le_4317_14677`. Both results are assembled in `PolarProjectionMain` from proved arithmetic, geometric and counting estimates. The repository README gives the correspondence between the other results and their Lean declarations.

The affine-curve theorem (24) is imported from the separate `ccdn-linear-growth` library at revision `9ade620`. That development proves the plane estimate, degree-preserving projection and box rescaling, using the same affine Hilbert degree as this paper. The present repository proves the required monic Hilbert irreducibility estimate and its application to the actual branch-label family. Thus neither counting input is an assumed proposition in the final polar bound.

The separate logarithmic refinement `P_le_loglog` retains two explicit hypotheses: the short-sum estimate for the Erdős–Hooley function from [9], over intervals $(x, x+y]$ with $y \geq x^{1/4}$, and the mean-value estimate from [2]. Their statements include the constants and thresholds; their proofs are not formalized here.

The proof library contains no `sorry` and uses no `native_decide`. Its axiom audit permits only `propext`, `Classical.choice` and `Quot.sound`, the standard axioms used by Mathlib. Fifteen public statements are compared with independent Mathlib-only specifications, and their proofs are replayed by Lean, NanoDa and con-ron. The specification file `Challenge.lean` deliberately contains theorem holes; it is separate from the proved solution. Build instructions, the full theorem map, and the verification commands are recorded in the README.

The divisor estimate uses `DivisorPowerBound.lean` from `openai/math` [10] under Apache-2.0, with unchanged proof bodies and adapted imports and module visibility. The repository records the source revision and the provenance of its other dependencies.

13. Limits of the method

The polar refinement brings the exponent close to 5/17, but does not remove that threshold. The current local estimates still contain the X^5 pattern contribution from bounded cuts and

conics, the earlier exceptional-pattern contribution, and the windows where the line-exclusion condition (19) fails. Each leads to exponent $5/17$, independently of the coordinate powers m, n. The new sieve and geometric exceptional locus fit below this threshold.

The improved graph degree gives

$$\beta_* - \frac{5}{17} = 0.00001600826\ldots.$$

Increasing the graph degree further can reduce this gap. Going below $5/17$ requires an improvement in the other counting contributions, or a different way to combine the information from successive orbit states. This is a limitation of the present estimates, not a lower bound for the Pierce iteration.

The earlier $3/10$ argument has a similar barrier. Even with no divisor loss, its two window bounds

$$T \ll a^{1/2}A^{-1/3}, \qquad T \ll A^{1/2}$$

balance at $A \asymp a^{3/5}$ and permit $T \asymp a^{3/10}$. Better constants or short-interval divisor estimates alone do not change this exponent. The higher invariants and fixed-dividend equations supply restrictions on configurations that this elementary balance misses.

The linear and quadratic pattern relations alone still admit families of solutions. For example, take the Newton coefficients

$$(\alpha, -3\alpha - 1, 3\alpha + 3, -\alpha - 3, 1)$$

at every state, with coefficient polynomial $(z-1)^3(z-\alpha)$, and constant increments $k_j = t$. All the linear and quadratic relations hold for every t. The remainder equations exclude such a free family; this is why retaining the fixed dividend is essential to the curve construction.

Nor do the curvatures necessarily decrease. For $a = 13242$, the orbit of 255 begins

$$255,\ 237,\ 207,\ 201,\ 177,\ 144,\ 138,\ 132, \ldots$$

with quotients $51, 55, 63, 65, 74, 91, 95, 100, \ldots$ and first five curvatures $24, 12, 6, 111, 30$. The corresponding blocks all lie within a factor of two. The rise bound gives one constraint on the order of the increments, but the present argument does not capture all compatibility conditions along an entire orbit. Shallit's conjectured $O((\log a)^2)$ bound remains open.

Appendix A. Conic components

In this appendix we prove Proposition 8.9. Let C be a nonexceptional component of degree two. A component that is not retained already contributes at most $404 \cdot 2 = 808$ starts, by Lemma 8.2. We may therefore assume that the component C is retained. If it contains at most two free-node triples of good starts, there is nothing to prove. In the remaining case we first construct a parameter whose value is an integer at every such triple.

Lemma A.1 (A bounded integral parameter)**.** *Let C be a retained nonexceptional degree-two component containing the free-node triples of at least three good starts. After permuting the three free-node coordinates to (x, y, z), there are an integer $\Delta_c \neq 0$ and polynomials $\xi_0, \xi_1, \xi_2 \in \mathbb{Z}[t]$ such that*

$$\deg \xi_j \leq 2, \qquad |\Delta_c|,\ \max_{j,k} |[t^k]\xi_j| \leq (aX)^{O(1)}.$$

The map

$$t \longmapsto \big(\xi_0(t), \xi_1(t), \xi_2(t)\big)/\Delta_c$$

parametrizes C over $\mathbb{Q}$. Its inverse takes an integer value at every integer point of C, and is injective on those points.

Proof. Every free node of a good start has absolute value at most a. Three distinct such triples admit an integer plane

$$\lambda_z z + \lambda_x x + \lambda_y y + \lambda_1 = 0, \qquad |\lambda_z|, |\lambda_x|, |\lambda_y| \le 8a^2, \qquad |\lambda_1| \le 24a^3.$$

For example, take a cross product of two differences to obtain the normal; if that cross product vanishes, a nonzero perpendicular vector with the same bounds suffices. Put $K_{\mathrm{p}} = 24a^3$. A plane not containing an integral curve of degree two meets it in at most two distinct points. Thus this plane contains C. Choose the eliminated coordinate z with $\lambda_z \neq 0$.

The six rational pattern coordinates (10) have common denominator $N_0 N_1 \neq 0$ and respective integer numerators

$$N_1^2,\ u_1 N_1,\ u_2 N_1,\ N_2 N_0,\ v_1 N_0,\ v_2 N_0,$$

each of absolute value at most X^2. Expanding the quadratic f, its homogeneous part is a diagonal quadratic plus the square of a linear form. Its three diagonal numerators have denominator $N_0 N_1$, and its three linear numerators have denominator $(N_0 N_1)^2$; all six numerators have absolute value at most $12X^4$. Substitute $z = -(\lambda_x x + \lambda_y y + \lambda_1)/\lambda_z$ and multiply by $(N_0 N_1 \lambda_z)^2$. The resulting plane equation is

$$\alpha x^2 + \beta xy + \gamma_c y^2 + \ell_x x + \ell_y y + \ell_0 = 0, \tag{38}$$

with integer coefficients of absolute value at most $K = 10(12X^4)^2 K_{\mathrm{p}}^2$. Indeed, each coefficient is a sum of products of two quantities bounded by $12X^4$ and two plane coefficients bounded by K_{p}; expansion gives the stated factor 10. The restriction of f to the plane is nonzero: otherwise the plane would be a component of the irreducible pattern quadric. It has degree two because it contains C, and hence defines C in that plane.

The pullback of the hyperplane at infinity to the normalization of C has degree two. Corollary 8.4 says that each of its positive multiplicities is even. Counting residue degrees, there is therefore just one branch over infinity, of multiplicity two and residue degree one. Consequently the leading binary quadratic in (38) is a nonzero rational scalar times the square of a rational linear form. Indeed, the normalization map is finite and surjective, so the projective conic meets the line at infinity only at the image of this rational point. Its leading binary quadratic therefore has a single rational zero, of multiplicity two. Interchanging x and y if necessary gives

$$\alpha \neq 0, \qquad \beta^2 - 4\alpha\gamma_c = 0.$$

Set

$$\kappa_c = 2\beta\ell_x - 4\alpha\ell_y, \qquad \epsilon_c = \ell_x^2 - 4\alpha\ell_0, \qquad t = 2\alpha x + \beta y + \ell_x.$$

Completing the square in (38) gives

$$t^2 = \kappa_c y + \epsilon_c. \tag{39}$$

Here $\kappa_c \neq 0$. If it were zero, a rational point of C with parameter t_0 would give $\epsilon_c = t_0^2$, and the equation would factor as $(t - t_0)(t + t_0) = 0$. Integrality would then place C in an affine line, contradicting its degree.

We can therefore solve explicitly:

$$y = \frac{t^2 - \epsilon_c}{\kappa_c}, \qquad x = \frac{t - \beta y - \ell_x}{2\alpha}, \qquad z = -\frac{\lambda_x x + \lambda_y y + \lambda_1}{\lambda_z}.$$

Substitution verifies the conic and plane equations for every rational t, and the displayed linear expression in x, y recovers t. In particular, t is an integer whenever (x, y, z) is an integer point.

Distinct points have distinct parameters. Taking $\Delta_c = 2\alpha\kappa_c\lambda_z$ clears all three denominators. The inequalities

$$|\kappa_c| \le 6K^2, \qquad |\epsilon_c| \le 5K^2, \qquad |\beta\epsilon_c - \ell_x\kappa_c| \le 11K^3$$

give $|\Delta_c| \le 12K_{\mathrm{p}}K^3$ and the coefficient bounds $33K_{\mathrm{p}}K^3$ for the three resulting quadratic numerators. For good starts one also has $|t| \le 4Ka$, although the counting argument below does not need a parameter interval. □

Lemma A.2 (Properness and coefficient height)**.** *In the parameter of Lemma A.1, write*

$$E_0(t) = \frac{F_E(t)}{G_E(t)}, \qquad F_E, G_E \in \mathbb{Z}[t], \qquad \gcd_{\mathbb{Q}[t]}(F_E, G_E) = 1.$$

The polynomials may be chosen so that

$$\deg G_E - \deg F_E = 3, \qquad \max(\deg F_E, \deg G_E) \le D_0, \qquad \max_k\big(|[t^k]F_E|, |[t^k]G_E|\big) \le C_0 a^{k_0},$$

for absolute constants D_0, C_0, k_0. In particular, G_E is nonconstant. At the parameter t of every good start, $G_E(t) \neq 0$ and $F_E(t)/G_E(t) = I_3(i)$ is an integer.

Proof. The explicit parametrization identifies C with the affine line and its smooth projective model with $\mathbb{P}^1$. Its point at infinity has hyperplane multiplicity two. Lemma 8.6 therefore gives

$$\operatorname{ord}_{t=\infty} E_0 = 3.$$

The function is nonzero, since it is positive at a good start. For a reduced rational function on $\mathbb{P}^1$, its order at infinity is $\deg G_E - \deg F_E$, proving properness.

We next bound the coefficients uniformly in the component, including after cancellation. Retain $K_{\mathrm{p}} = 24a^3$ and $K = 10(12X^4)^2K_{\mathrm{p}}^2$ from the preceding proof, and define

$$K_{\mathrm{h}} = 1 + a + X + 33K_{\mathrm{p}}K^3.$$

The six rational pattern coordinates, the three coordinate polynomials ξ_j/Δ_c, and the fixed dividend a have the common nonzero denominator $N_0N_1\Delta_c$. Their ten numerator polynomials have degree at most two and coefficient height at most K_{h}^4; the common denominator has absolute value at most K_{h}^4 as well. For instance, the coordinate numerators are $N_0N_1\xi_j$, while the numerator for a is $aN_0N_1\Delta_c$.

The cofactor formula defining E_0 is a quotient of two fixed polynomials in these ten inputs, with fixed rational coefficients. Clear those fixed coefficients once, then substitute the ten polynomials and clear a fixed power of their common denominator. This gives integer polynomials $\tilde{n}, \tilde{d}$, both nonzero, of bounded degree and coefficient height at most CK_{h}^r, where the degree bound, C and r are absolute. This assertion follows directly from polynomial multiplication: both the number of factors and the number of summands are fixed by the displayed cofactor formula.

Choose a primitive $\Pi \in \mathbb{Z}[t]$ representing $\gcd_{\mathbb{Q}[t]}(\tilde{n}, \tilde{d})$. By Gauss's lemma, Π divides both $\tilde{n}$ and $\tilde{d}$ in $\mathbb{Z}[t]$. Thus $F_E = \tilde{n}/\Pi$ and $G_E = \tilde{d}/\Pi$ are integer polynomials with the same ratio, coprime over $\mathbb{Q}$, and with degrees no larger than those of $\tilde{n}, \tilde{d}$. For a nonzero integer polynomial of degree at most D_0 and coefficient height $\mathcal{H}$, every integer polynomial factor has coefficient height at most $2^{D_0}(D_0+1)\mathcal{H}$. To see this, the Mahler measure of an integer cofactor is at least one, the original polynomial's Mahler measure is at most $(D_0+1)\mathcal{H}$, and each coefficient of a factor is bounded by 2^{D_0} times its Mahler measure. Thus cancellation preserves a fixed polynomial bound in K_{h}.

Finally, $K_{\mathrm{p}} = 24a^3$ and $X \le 10^{16}a^7$ show that K_{h} is bounded by a fixed power of a. Absorbing the fixed exponents and constants gives the asserted $C_0a^{k_0}$ bound. The original

reconstruction denominator is nonzero at every good start. Its cleared polynomial $\tilde{d}$ is therefore nonzero at the corresponding parameter, and so is its factor G_E. The reconstruction identity then gives $F_E(t)/G_E(t) = I_3(i) \in \mathbb{Z}$. □

Lemma A.3 (Integer values of a rational function)**.** *Let $F_E, G_E \in \mathbb{Z}[t]$ be coprime over $\mathbb{Q}$, of degrees at most D_0, with G_E nonconstant and coefficient heights at most $C_0 a^{k_0}$. For every $\eta > 0$, the number of integers t such that $G_E(t) \neq 0$ and $F_E(t)/G_E(t) \in \mathbb{Z}$ is $O_\eta(a^\eta)$. The implied constant depends only on η, D_0, C_0, k_0.*

Proof. The integer resultant $R = \operatorname{Res}(F_E, G_E)$ is nonzero. The adjugate of the Sylvester matrix gives $U, V \in \mathbb{Z}[t]$ with $UF_E + VG_E = R$. At any parameter in question, $G_E(t) \mid F_E(t)$, and hence $G_E(t) \mid R$. There are $2\tau(|R|)$ possible signed values of $G_E(t)$. For each such value v, the nonzero polynomial $G_E - v$ has at most $\deg G_E \le D_0$ roots. Thus the number of parameters is at most

$$2D_0\tau(|R|).$$

The Sylvester matrix has size at most $2D_0$, and its entries are bounded by $C_0 a^{k_0}$. Expanding its determinant gives

$$1 \le |R| \le (2D_0)!\big(C_0 a^{k_0}\big)^{2D_0}.$$

The divisor estimate $\tau(n) \ll_\varepsilon n^\varepsilon$, with ε chosen in terms of η and the fixed degree and height exponents, proves the claim. □

Proof of Proposition 8.9. Apply the last lemma to the reconstructed invariant E_0. The parameter is injective on free-node triples, and reconstruction of the strictly increasing q_i makes those triples injective in the good starts. Each retained nonexceptional conic therefore contributes $O_\eta(a^\eta)$ starts. The components with at most two free-node triples of good starts and the nonretained components were already bounded absolutely. Apply the conic estimate with $\eta/2$ and the pattern estimate $O_{\eta/2}(a^{\eta/2}X^5)$. Since there are at most twenty-four components per pattern, all nonexceptional conics together contribute $O_\eta(a^\eta X^5)$. □

Appendix B. The upper bound for graph degrees

Proof of the upper bound in Lemma 8.12. Here only total polynomial degrees matter, not the heights of the pattern-dependent coefficients. Write the cofactor formula for the reconstructed invariant as $E_0 = E_{\mathrm{n}}/E_{\mathrm{d}}$, as in the proof of Lemma 8.2. Each entry of the moment matrix has degree at most four in the nodes. Expanding its 8×8 cofactors gives, in particular, the uniform bound $\deg c_j \le 40$, so $\deg E_{\mathrm{n}}, \deg E_{\mathrm{d}} \le 240$. Since H_3, H_3' have degree at most three and the gaps have degree at most one, the three curvature coordinates have a common denominator

$$\Omega = E_{\mathrm{d}} H_3 k_1, \qquad \deg \Omega \le 244,$$

and numerators $\mathcal{N}_0, \mathcal{N}_1, \mathcal{N}_2$ of degree at most 244. The numerator for J with this denominator is $\mathcal{N}_3 = J\Omega$, of degree at most 247. The denominator is nonzero on C, since it is nonzero at a good start.

Put $\bar{m} = m + n$. Represent the powered coordinates with denominator $\Omega^{\bar{m}}$ and numerators

$$\mathcal{N}_j^m \Omega^{\bar{m}-m} \quad (0 \le j \le 2), \qquad \mathcal{N}_3^n \Omega^{\bar{m}-n}.$$

All these numerators have degree at most $247\bar{m}$. The original node coordinates, written with the same denominator, have numerators of degree at most $244\bar{m}+1$. It follows that multiplying a polynomial of degree at most t in the graph coordinates by $\Omega^{\bar{m}t}$ gives a polynomial in the original nodes of degree at most $(247\bar{m}+1)t$. This operation is injective on the corresponding

spaces of functions, because Ω is nonzero in $\mathbb{Q}(C)$. Comparing their eventual affine Hilbert functions gives

$$\delta t + O(1) \le d(247\bar{m} + 1)t + O(1),$$

and hence the asserted upper bound. In the unpowered case use Ω directly: the four new numerators have degree at most 247, and the original coordinates have numerators of degree at most 245. The same comparison gives $\delta \le 247d$. □

Appendix C. A reduced basis in dimension three

The constant $64/27 = (4/3)(16/9)$ in Lemma 9.3 comes from two successive size reductions. We give the elementary argument for completeness.

Proof of Lemma 9.3. Let V be the least nonzero $|\det|$ of a triple in Λ; the triple $(\kappa\mathbf{e}_1, \kappa\mathbf{e}_2, \kappa\mathbf{e}_3)$ gives $V \le \kappa^3$. A triple $(\beta_1, \beta_2, \beta_3)$ with $|\det| = V$ is a basis: if not, some $x \in \Lambda$ has a Cramer coordinate that is not an integer, say the first, $\det(x, \beta_2, \beta_3)/\det(\beta_1, \beta_2, \beta_3)$, and subtracting the integer part of that coordinate times β_1 from x gives a triple with smaller nonzero determinant. Among all bases (all have $|\det| = V$) minimize $g_1 = |\beta_1|^2$, and then the Gram determinant $g_2 = g_1|\beta_2|^2 - \langle\beta_1, \beta_2\rangle^2$. Put $g_3 = V^2$.

Choose an integer z_1 with $|2(\langle\beta_1, \beta_2\rangle + z_1 g_1)| \le g_1$. Then $(\beta_2 + z_1\beta_1, \beta_1, \beta_3)$ is a basis, so $g_1 \le |\beta_2 + z_1\beta_1|^2$, and $g_1|\beta_2 + z_1\beta_1|^2 = g_2 + (\langle\beta_1, \beta_2\rangle + z_1 g_1)^2$. Hence $3g_1^2 \le 4g_2$.

Let $\langle x, x'\rangle_1 = g_1\langle x, x'\rangle - \langle\beta_1, x\rangle\langle\beta_1, x'\rangle$, so that $\langle\beta_2, \beta_2\rangle_1 = g_2$, and Sylvester's identity gives $\langle\beta_2, \beta_2\rangle_1\langle\beta_3, \beta_3\rangle_1 - \langle\beta_2, \beta_3\rangle_1^2 = g_1 g_3$. Choose an integer z_2 with $|2(\langle\beta_2, \beta_3\rangle_1 + z_2 g_2)| \le g_2$ and put $y = \beta_3 + z_2\beta_2$. Then (β_1, y, β_2) is a basis with first vector β_1, so $g_2 \le g_1|y|^2 - \langle\beta_1, y\rangle^2 = \langle y, y\rangle_1$, while $g_2\langle y, y\rangle_1 = (\langle\beta_2, \beta_3\rangle_1 + z_2 g_2)^2 + g_1 g_3$. Hence $3g_2^2 \le 4g_1 g_3$.

Finally choose an integer z_3 with $|2(\langle\beta_1, y\rangle + z_3 g_1)| \le g_1$, and use the basis β_1, $\beta_2' = \beta_2 + z_1\beta_1$, $\beta_3' = y + z_3\beta_1$. The identities above give $3g_1|\beta_2'|^2 \le 4g_2$ and $4g_1 g_2|\beta_3'|^2 \le g_2^2 + 4g_1 g_3 + g_1^2 g_2$. With the two inequalities this gives $9g_2|\beta_3'|^2 \le 16 g_3$. Since $|\beta_1|^2 = g_1$, the product of the bounds $|\beta_2'|^2 \le \frac{4}{3} g_2/g_1$ and $|\beta_3'|^2 \le \frac{16}{9} g_3/g_2$ is the claim for the basis $\beta_1, \beta_2', \beta_3'$; rename β_2', β_3' as β_2, β_3. Replacing β_1 by $-\beta_1$ if necessary makes the determinant positive. For the coordinates, $\xi_1 V = \det(x, \beta_2, \beta_3)$, and Hadamard's inequality gives $\det(x, \beta_2, \beta_3)^2 \le |x|^2|\beta_2|^2|\beta_3|^2$, so $27|\beta_1|^2\xi_1^2 V^2 \le 64 V^2|x|^2$; likewise for ξ_2 and ξ_3. □

Appendix D. Certificates for the polar bound

D.1. **The quartic initial form at the base point.** At $(\rho, u, v, \rho', u', v') = (2, 5, -3, 3, 6, -5)$ the polar base point is $x_* = (-1/7, -3/28, -1/2)$ and $\nabla f(x_*) = (-4, 9, -5)$. Use tangent coordinates (s, t) with first-order displacement

$$\delta x = ((9s - 5t)/4, s, t).$$

Thus the two axes used in Lemma 10.2 have tangent vectors $(9/4, 1, 0)$ and $(-5/4, 0, 1)$. Direct substitution in the Newton quartic, its remainder and the norm polynomial gives

$$\begin{aligned}
P_{4,*}(\xi) &= \xi^4 - \frac{20}{7}\xi^3 + \frac{757}{112}\xi^2 - \frac{669}{112}\xi + 2,\\
R_1(\xi; s, t) &= \frac{77s + 7t}{14} + \frac{-56s - 124t}{14}\xi + \frac{-45s + 157t}{14}\xi^2 + \frac{28s - 28t}{14}\xi^3,\\
\Psi_1(s, t) &= \frac{3s + 85t}{28}.
\end{aligned}$$

The resultant is

$$N_4(s, t) = \frac{32233977s^4 - 95047956s^3t + 147873750s^2t^2 - 111199572st^3 + 31702169t^4}{6272}.$$

In particular,

$$N_4(1,0)\Psi_1(0,1)^4 - N_4(0,1)\Psi_1(1,0)^4 = \frac{26291112152499}{60236288}.$$

These identities specify the frame and first derivatives needed to reproduce the local certificate. The realizing matrix in Lemma 10.2 certifies a nonzero polynomial on the decomposable cone; no orbit with these parameters is required.

D.2. **The irreducible sextic.** The specialization $E_\gamma = 24E_0$ at $(\gamma_1, \gamma_2, \gamma_3) = (-7, 8, -4)$ reduces modulo 71 to a nonzero scalar multiple of

$$q(T) = T^6 + 20T^5 - 14T^4 + 15T^3 - 28T^2 + 5T - 4.$$

More precisely, $E_0 = 57q$ in $\mathbb{F}_{71}[T]$. The following certificates prove that q is irreducible. Repeated squaring modulo q gives

$$T^{71^2} \equiv r_2(T) := 36T^5 + 48T^4 + 49T^3 + 53T^2 + 36T + 42,$$
$$T^{71^3} \equiv r_3(T) := 37T^5 + 22T^4 + 60T^3 + 13T^2 + 66T + 35.$$

For checking the iteration one may first use

$$T^{71} \equiv 14T^5 + 33T^4 + 31T^3 + 34T^2 + 48T \pmod{q}$$

and raise each remainder to the 71st power. Define

$$U_2 = 20T^4 + 34T^3 + 31T^2 + 13T + 24,$$
$$V_2 = 31T^5 + 61T^4 + 42T^3 + 61T^2 + 39T + 4,$$
$$U_3 = 25T^4 + 11T^3 + 53T^2 + 59T + 48,$$
$$V_3 = 7T^5 + 30T^4 + 65T^3 + 64T^2 + 25T + 40.$$

Direct multiplication in $\mathbb{F}_{71}[T]$ verifies

$$U_2 q + V_2(r_2 - T) = 1, \qquad U_3 q + V_3(r_3 - T) = 1.$$

Thus q has no irreducible factor of degree one or two, since such a factor would divide $T^{71^2} - T$, and none of degree three, since it would divide $T^{71^3} - T$. Every reducible sextic has an irreducible factor of degree at most three, so q is irreducible. Reduction modulo 71 preserves degree, and Gauss's lemma proves E_0 irreducible over $\mathbb{Q}$.

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References

[1] O. Baraskar and I. Vukusic, *Bounds for sets of remainders*, arXiv:2508.20853 (2025).

[2] R. de la Bretèche and G. Tenenbaum, *Note on the mean value of the Erdős–Hooley Delta-function*, Acta Arith. **219** (2025), no. 4, 379–394.

[3] W. Castryck, R. Cluckers, P. Dittmann and K. H. Nguyen, *The dimension growth conjecture, polynomial in the degree and without logarithmic factors*, Algebra Number Theory **14** (2020), no. 8, 2261–2294, doi:10.2140/ant.2020.14.2261.

[4] R. Cluckers, P. Dèbes, Y. I. Hendel, K. H. Nguyen and F. Vermeulen, *Improvements on dimension growth results and effective Hilbert's irreducibility theorem*, Forum Math. Sigma **13** (2025), e153, doi:10.1017/fms.2025.10096.

[5] Z. Chase and M. Pandey, *On the length of Pierce expansions*, arXiv:2211.08374 (2022).

[6] P. Erdős and J. O. Shallit, *New bounds on the length of finite Pierce and Engel series*, Sém. Théor. Nombres Bordeaux (2) **3** (1991), 43–53.

[7] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[8] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 367–381.

[9] M. Nair and G. Tenenbaum, *Short sums of certain arithmetic functions*, Acta Math. **180** (1998), 119–144.

[10] OpenAI, `openai/math`, `https://github.com/openai/math`.

[11] J. O. Shallit, *A discrete iteration related to Pierce expansions*, Open Problem Garden, `https://www.openproblemgarden.org/op/a_discrete_iteration_related_to_pierce_expansions`.

[12] T. A. Pierce, *On an algorithm and its use in approximating roots of algebraic equations*, Amer. Math. Monthly **36** (1929), 523–525.

[13] J. O. Shallit, *Metric theory of Pierce expansions*, Fibonacci Quart. **24** (1986), 22–40.

[14] J. O. Shallit, *Talks*, `https://cs.uwaterloo.ca/~shallit/talks.html` (accessed October 2026).

Email address: `jason@jasondavies.com`